



Master's Thesis

**Nonlinear Conjugate Gradient with Perturbed
Heavy-Ball Saddle Escape:
Complexity and Local Convergence**

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Abstract

Nonlinear conjugate gradient (NCG) methods are widely used for large-scale smooth optimization, where they cost no more per iteration than gradient descent and often converge much faster. Their theoretical guarantees, however, are limited in the nonconvex setting: while a safeguarded variant is known to converge to an approximate first-order stationary point, it cannot guarantee escape from saddle points. Meanwhile, other first-order methods, such as gradient descent or accelerated gradient, can be endowed with such guarantees by adding small perturbations to the iterates when those come close to a stationary point.

In this thesis, we propose an algorithm that utilizes NCG steps for descent and transitions to a fixed number of perturbed heavy-ball iterations whenever the gradient falls below a specified threshold to certify saddle points avoidance. We prove that it returns an approximate second-order stationary point with high probability under standard smoothness assumptions, at a cost comparable to perturbed gradient descent. To our knowledge, this is the first such guarantee for a method based on NCG.

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Chapter 1

Introduction

Nonconvex optimization is the backbone of training machine learning models and many data science tasks can be modeled as nonconvex problems. Typical examples include matrix completion [26], robust PCA [45] and phase retrieval [4]. In this work, we consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \quad (1.1)$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice Lipschitz continuously differentiable nonconvex function. In nonconvex optimization, it is typically intractable to certify convergence to a minimum, either local or global, without stronger assumptions. Moreover, contrary to convex optimization, it is not sufficient either to certify convergence to an ε -first-order stationary point (FOSP), that is, a point x such that $\|\nabla f(x)\| \leq \varepsilon$, since such a point could be a local maximum or a saddle point. Therefore, we are interested in finding an $(\varepsilon_g, \varepsilon_H)$ -second-order stationary point (SOSP), that is, a point x such that

$$\|\nabla f(x)\| \leq \varepsilon_g \quad \text{and} \quad \lambda_{\min}(\nabla^2 f(x)) \geq -\varepsilon_H, \quad (1.2)$$

where $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^n$. In some instances, finding an $(\varepsilon_g, \varepsilon_H)$ -SOSP is equivalent to finding an approximate global minimum. Many problems in machine learning have benign landscapes [3, 17, 34, 58, 59]: every local minimum is global, and every saddle point is strict, that is, it admits a direction of sufficiently negative curvature. On such landscapes, a point satisfying (1.2) with small enough tolerances lies close to a global minimum. It has been shown that the main challenge in finding an $(\varepsilon_g, \varepsilon_H)$ -SOSP in nonconvex optimization is attributable to escaping saddle points [13].

The best methods to avoid such points are the so-called second-order methods, that is, methods using the Hessian to navigate through the optimization landscape, such as trust-region [8] and cubic regularization [5, 44] methods. It has been established that those methods have the best complexity guarantees for finding an $(\varepsilon_g, \varepsilon_H)$ -SOSP of $\mathcal{O}(\max\{\varepsilon_g^{-3/2}, \varepsilon_H^{-3}\})$ [6, 44]. However, those guarantees come with the significant cost of Hessian computation at each iteration, which is not tractable in high dimension [46]. Although some methods do not directly compute the Hessian and access negative curvature information through Hessian-vector products [35, 51, 57] or through gradient evaluations alone [1, 64], they still cost more than first-order methods.

The baseline first-order minimization algorithm is gradient descent (GD) defined as

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k), \quad (1.3)$$

where α_k is the stepsize at step k . GD reaches an ε -FOSP in $\mathcal{O}(\varepsilon^{-2})$ [43]. It is known that other first-order methods have better guarantees.

Momentum techniques are first-order methods that re-use previous information at each step to accelerate the optimization process, while having the same cost of one gradient evaluation at

each iteration as other first-order methods. A standard momentum algorithm is the Heavy-Ball (HB) [53] method:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k) + \gamma (x_k - x_{k-1}), \quad (1.4)$$

where γ is a hyperparameter controlling the amount of momentum included at each iteration, and α_k is the stepsize at step k .

The other classical momentum method is Nesterov’s accelerated gradient descent (AGD) [42], which evaluates the gradient at an extrapolated point rather than at the current iterate:

$$y_k = x_k + \mu_k (x_k - x_{k-1}), \quad x_{k+1} = y_k - \alpha_k \nabla f(y_k), \quad (1.5)$$

where μ_k is the extrapolation coefficient. On smooth convex functions, AGD reaches a better convergence rate than HB.

Nonlinear conjugate gradient (NCG) methods [23, 54] are another family of momentum techniques [63]. They have been found to be very effective for nonlinear smooth large-scale problems [46]. An iteration of NCG reads as

$$x_{k+1} = x_k + \alpha_k d_k, \quad d_{k+1} = -\nabla f(x_{k+1}) + \beta_{k+1} d_k, \quad (1.6)$$

where d_k is the search direction, $d_0 = -\nabla f(x_0)$, and α_k is a stepsize obtained by Armijo backtracking line search [11]. Individual NCG methods are distinguished by the choice of β_{k+1} [15, 25, 52] which is typically a nonlinear combination of gradients at steps k and $k+1$. This nonlinearity is what makes these methods hard to analyze: d_{k+1} may not be a descent direction, and no complexity guarantee is available for NCG in this form.

A restarted variant of NCG from [7] imposes a safeguard on the conjugate direction. At each iteration, it tests the direction and chooses whether to keep it or take a GD step instead. This yields the first complexity guarantee for a NCG method in the nonconvex setting, for reaching an ε_g -FOSP.

The main challenge in going from a FOSP to a SOSP is to verify that the FOSP is not a saddle point. It has been shown that GD escapes strict saddle points [37], but the guarantee is asymptotic, and there exist smooth functions on which GD takes exponentially many iterations to escape [14]. Moreover, this proof technique does not apply to NCG (see Section 2.6). Later, a line of work initiated by [29] showed that some perturbed first-order methods reach an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability and with explicit complexity: perturbed gradient descent (PGD) [29], and perturbed accelerated gradient descent (PAGD) [30].

Every subsequent perturbed method with complexity guarantees is built on either GD or AGD. More generally, no complexity result for the convergence of a NCG method to an $(\varepsilon_g, \varepsilon_H)$ -SOSP is available, whether obtained by perturbation or not.

1.1 Contributions

We design an algorithm in two phases. While the gradient is large, it takes safeguarded NCG steps with the step size chosen by Armijo backtracking line search. When the gradient falls below ε_g , the iterate is a candidate SOSP: the algorithm perturbs it and runs a fixed number of HB steps. If those steps decrease f enough, the candidate is not an $(\varepsilon_g, \varepsilon_H)$ -SOSP and the NCG phase resumes; otherwise the candidate is an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability, and the algorithm stops.

The two-phase structure is well suited to the proof technique of [29, 30], which compares two runs of the method launched from nearby points, and requires both to use the same stepsize and momentum coefficient at every step. NCG provides neither: α_k comes from a line search and β_k from a nonlinear combination of the gradients. HB provides both, since its stepsize and momentum coefficient can be held fixed.

The main contribution of this work is a second-order complexity guarantee for this method. Theorem 1 shows that Algorithm 2 returns an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability under gradient- and Hessian-Lipschitz continuity, after at most

$$\tilde{\mathcal{O}}(\varepsilon_g^{-2} + \varepsilon_g^{-\max\{1+p, 2(1+p-q)\}} + \varepsilon_H^{-4})$$

iterations. Here p and q are hyperparameters of the restart test of NCG, typically between 0 and 1, and $\tilde{\mathcal{O}}$ hides polylogarithmic dependence in the dimension. Within the descent phase, we have two rates in the complexity as we must account for both restarted and non-restarted iterations. ε_g^{-2} , which is the GD rate, is charged only to steps where restart happens, and the second rate to steps where NCG iterations are performed. Hence the overall bound is similar to GD in the worst case, but can be better in a favourable case with p and q chosen such that the exponent of the second rate is lower than 2, and no restarts happen. The bound of the escape phase matches that of PGD [29], but our escape analysis borrows arguments from both the PGD and the PAGD [30] analyses.

To support of theoretical findings, we implement and test our method on two strict-saddle instances. The first finding is that the theoretical constants we find in this work are too conservative to be applied empirically. However we propose a solution for implementation. Then, we measure convergence speed in number of iterations to reach an $(\varepsilon_g, \varepsilon_H)$ -SOSP. Our algorithm is faster than PGD on both instances and faster than PAGD on only one out of the two. We also measure the number of evaluations (function and gradient) to reach an $(\varepsilon_g, \varepsilon_H)$ -SOSP. With this metric, our algorithm beats both PGD and PAGD on both instances.

Finally, under the additional assumption that f satisfies the strict saddle property, we show that the stationary point returned by Algorithm 2 lies in a strongly convex basin, and we establish local convergence with complexity.

1.2 Organisation

The rest of this work is organized as follows. Chapter 2 surveys the related work. Chapter 3 presents the algorithm and the main complexity result (Section 3.1), then proves it: Section 3.2 analyses the descent phase, Section 3.3 the escape phase, which is the heart of this work, and Section 3.4 combines the two into the proof of Theorem 1. Chapter 4 reports the numerical experiments, and Chapter 5 adds a local convergence statement under an additional landscape assumption. Chapter 6 concludes and discusses future work.

1.3 Notations

Throughout this work, we denote $g_k := \nabla f(x_k)$. We write f_{low} for a lower bound on f and $\Delta_f := f(x_0) - f_{\text{low}}$ for the initial optimality gap. We use the global index k for the main iteration counter of Algorithm 2. The perturbed HB steps are counted separately, by a local index $t \in \{0, \dots, T\}$, and the corresponding iterates are written $\hat{x}_t$; quantities derived from them, such as g_t and d_t , carry the same local index (see Section 3.1). We write $B_x(R) := \{y : \|y - x\| \leq R\}$ for the closed ball of centre x and radius R in $\mathbb{R}^n$; when a ball in another dimension is needed, we indicate it by a superscript, as in $B_x^{(n-1)}(R)$. We use $\mathcal{O}$, Ω and Θ in the usual sense of upper, lower and two-sided bounds up to constants, and $\tilde{\mathcal{O}}$ to hide polylogarithmic dependence.

Chapter 2

Literature Review

This literature review reflects my development during this master’s thesis. The first four sections are in chronological order of my discoveries and work. I first learnt about strict saddle functions and why it is useful, then I modified a hybrid algorithm to use NCG and negative curvature descent and did my first complexity analysis presented in Chapter 5. Afterwards, we decided to work on first-order saddle avoidance, and I discovered the center stable manifold theorem, and tried really hard to make it work for NCG (I failed). Finally, we decided to take the perturbed road and I discovered the work of [29, 30] that I tried to adapt to NCG, eventually leading to the method presented in Chapter 3.

2.1 The strict saddle landscape

Chapter 1 called a landscape benign when every local minimum is global and every saddle is strict. That condition constrains only the critical points. Ge et al. [16] introduce an alternate notion characterizing benign nonconvex landscapes. A function (and, by extension, its landscape) is called strict saddle if at every point, either the gradient is large, or the Hessian has a direction of sufficiently negative curvature, or the point is already close to a local minimizer in whose neighbourhood f is strongly convex. Under that property, an $(\varepsilon_g, \varepsilon_H)$ -SOSP with small enough tolerances is guaranteed to lie near a local minimizer. Strict saddle landscapes arise in tensor decomposition [16], dictionary learning [58], phase retrieval [59], matrix sensing [3] and completion [17], and some neural networks [34]. There, an $(\varepsilon_g, \varepsilon_H)$ -SOSP lies near a global minimizer.

The property has also been exploited inside algorithms rather than only in their analysis: on Riemannian manifolds, [20] show that a trust-region method applied to strict saddle functions reaches an approximate local minimizer in a number of iterations depending only logarithmically on the accuracy parameter, an improvement over the bounds available for general nonconvex functions. Higher-order conditions have also been considered: [2] give an efficient algorithm converging to a third-order local optimum, which escapes degenerate saddles that the first two derivatives cannot distinguish from local minima, but show that finding a fourth-order local optimum is NP-hard. Such conditions require access to higher derivatives, and we restrict attention throughout to (1.2).

2.2 Escaping saddle points with curvature information

Second-order methods reach an $(\varepsilon_g, \varepsilon_H)$ -SOSP by exploiting negative curvature. The bound $\mathcal{O}(\max\{\varepsilon_g^{-3/2}, \varepsilon_H^{-3}\})$ quoted in Chapter 1 was established for cubic regularization of Newton’s method by Nesterov and Polyak [44], is attained by the adaptive framework of Cartis et al. [5], and is sharp for this class of methods [6]. Trust-region and cubic-regularization methods both

admit inexact subproblem solves accessing the Hessian only through matrix-vector products, by truncated conjugate gradient [57] for the trust-region subproblem and Lanczos iterations [35] for the smallest eigenvalue.

Even in that cheapest form, a second-order method queries curvature at every iteration. Hybrid methods reduce the number of queries by separating two regimes. While the gradient is large, they take inexpensive first-order steps; when it falls below ε_g , the iterate is a candidate SOSP and the method invokes a curvature oracle. A textbook presentation of this template, combining gradient descent with negative-curvature descent, is given in Section 9.3.1 of Wright and Ma [62]. Other work follow this template like [55] do with a line search, taking a Newton-like direction where the Hessian is positive definite and a negative-curvature direction otherwise. Royer et al. [56] replace the matrix factorizations by conjugate gradient and Lanczos iterations, so that the Hessian is accessed only through matrix-vector products, and Curtis et al. [10] obtain the analogous result in a trust-region framework. The oracle remains more expensive than a first-order step. In its cheapest Hessian-vector form, each query costs $\min\{n, O(\varepsilon_H^{-1/2})\}$ products, up to a factor logarithmic in ε_H and in the failure probability [56].

2.3 Deterministic avoidance of saddle points

Since second-order methods are expensive, it is natural to ask whether curvature information is needed at all to avoid saddle points, or merely useful. A line of work addresses this by casting an algorithm as a dynamical system $x_{k+1} = G(x_k)$ and applying the Center-Stable Manifold Theorem: when G is a local diffeomorphism, the set of initializations converging to a strict saddle is contained in a lower-dimensional manifold and therefore has Lebesgue measure zero. Lee et al. [37] first used this for gradient descent with a small constant step size, and Lee et al. [36] extended it to a wide variety of first-order methods, including manifold gradient descent, block coordinate descent and mirror descent, showing that neither second-order information nor randomness beyond the initialization is needed to avoid strict saddles.

These results have since been strengthened in two directions. On the hypotheses about f , Panageas and Piliouras [49] remove the requirement that critical points be isolated and allow non-globally-Lipschitz smoothness on forward-invariant convex subspaces. On the hypotheses about the algorithm, the theorem in its classical form applies only to *autonomous* systems, in which the same map is used at every iteration; this forces a constant step size and excludes the schedules used in practice. Panageas et al. [50] prove a stable manifold theorem for time non-homogeneous systems and obtain avoidance for gradient, mirror, manifold and proximal point descent with vanishing step sizes. Muşat and Boumal [40] treat a modified Armijo backtracking method with arbitrary initial step size, without assuming a Lipschitz gradient. In a subsequent work [41], the same authors establish a general Center-Stable Set Theorem for non-autonomous systems $x_{k+1} = G_k(x_k)$ covering vanishing step sizes without assuming Lipschitz gradients or isolated saddles. Parallel results exist in the Riemannian setting, for retraction-based stochastic methods [28] and for non-isolated saddle sets under projected gradient descent [27].

In the aforementioned literature, only asymptotic convergence guarantees are established, and complexity guarantees cannot be proven in general. Indeed, Du et al. [14] exhibit natural random initialization schemes and benign functions on which GD takes exponential time to escape, and show that PGD does not suffer this slowdown.

2.4 Perturbed first-order methods

The results of Section 2.3 guarantee that strict saddle points are avoided, but without complexity guarantees. To obtain such results only with first-order oracles, a second line of work adds randomness to the iteration itself.

Ge et al. [16] first showed that a stochastic gradient method with an isotropic perturbation added at every iteration converges to a point close to a local minimizer in polynomially many iterations, under the assumption that the objective satisfies the strict saddle property (Definition 1). Their rate is polynomial in the problem dimension. Jin et al [29] removed the landscape assumption and reduced that dependence to polylogarithmic: their PGD adds a uniform perturbation only when the gradient is small, and reaches an $(\varepsilon_g, \varepsilon_H)$ -SOSP of an arbitrary smooth nonconvex objective in $\tilde{\mathcal{O}}(\varepsilon_g^{-2})$ iterations, matching the rate of gradient descent to first-order stationarity up to logarithmic factors. The analysis rests on a characterization of the geometry around a saddle: the set of perturbations from which the method fails to escape is thin in the direction of most negative curvature, so a uniformly drawn perturbation misses it with high probability.

We recall PGD in Algorithm 1. The parameters are set from the Lipschitz constants, ε_g and

Algorithm 1 Perturbed gradient descent (PGD) [29]

Require: x_0 , stepsize η , perturbation radius r , gradient threshold g_{thres} , window length t_{thres} , decrease threshold f_{thres} .

```

1:  $t_{\text{noise}} \leftarrow -t_{\text{thres}} - 1$ 
2: for  $t = 0, 1, \dots$  do
3:   if  $\|\nabla f(x_t)\| \leq g_{\text{thres}}$  and  $t - t_{\text{noise}} > t_{\text{thres}}$  then
4:      $\tilde{x}_t \leftarrow x_t, \quad t_{\text{noise}} \leftarrow t$ 
5:      $x_t \leftarrow \tilde{x}_t + \xi_t, \quad \xi_t \sim \text{Unif}(B_0(r))$ 
6:   end if
7:   if  $t - t_{\text{noise}} = t_{\text{thres}}$  and  $f(x_t) - f(\tilde{x}_{t_{\text{noise}}}) > -f_{\text{thres}}$  then
8:     return  $\tilde{x}_{t_{\text{noise}}}$ 
9:   end if
10:   $x_{t+1} \leftarrow x_t - \eta \nabla f(x_t)$ 
11: end for
```

the failure probability. A perturbation is drawn when the gradient falls below g_{thres} and none has been drawn in the last t_{thres} iterations, that is, the current iteration is not already in an escape window. The perturbed point $\tilde{x}_{t_{\text{noise}}}$ is stored and t_{thres} gradient steps later the method compares f at the current iterate with $f(\tilde{x}_{t_{\text{noise}}})$. A decrease of at least f_{thres} means the escape succeeded, and the method continues; otherwise $\tilde{x}_{t_{\text{noise}}}$ is returned as an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability.

Jin et al. [30] extend the algorithm to AGD, obtaining an $(\varepsilon_g, \varepsilon_H)$ -SOSP in $\tilde{\mathcal{O}}(\varepsilon_g^{-7/4})$ iterations, the first single-loop accelerated method provably faster than GD in the general nonconvex setting. Their analysis introduces the *improve or localize* principle: along a stretch of iterations, either the objective decreases substantially or the iterates remain confined to a small ball.

Ma et al. [39] analyze GD in over-parameterized models and introduce infinitesimally perturbed gradient descent (IPGD). They show that infinitesimal perturbations suffice to escape saddle points while keeping the iterates in an implicit low-dimensional region, which yields guarantees for over-parameterized matrix sensing.

Analogous results hold on manifolds. Sun et al. [61] obtain an $\mathcal{O}(\varepsilon_g^{-2})$ rate for perturbed Riemannian gradient descent, and Criscitiello and Boumal [9] extend the analysis of Jin et al. [29] with a distinction between steps taken on the manifold and perturbed steps taken in a tangent space, reaching an SOSP in $\mathcal{O}((\log n)^4 \varepsilon_g^{-2})$ gradient queries and so recovering the near-dimension-free behaviour of the Euclidean case. All of these methods take GD, or a variant of it, as the underlying iteration.

2.5 The heavy-ball method in the nonconvex setting

The Heavy-ball (HB) method [53] was introduced for convex problems, where it converges at a linear rate to the global minimizer on smooth strongly convex quadratics. It remains under active study in that setting: [19] recently showed that on smooth strongly convex functions, no choice of parameters makes HB accelerated, closing a long-standing question on the method.

On nonconvex functions the objective is no longer monotonically decreasing along the iterates of HB, since the momentum term can make the step an ascent direction, so the argument underlying gradient methods is unavailable. Convergence is instead established through a Lyapunov function argument, as in [47, 65].

Momentum also changes the behaviour near saddle points. By the stable manifold argument of Section 2.3, HB can be shown to converge to a strict saddle only from a set of initializations of measure zero [48, 60]. O'Neill and Wright [48] further show that on a nonconvex quadratic, both HB and AGD move away from such a point faster than steepest descent. Their analysis must handle the fact that a momentum iteration depends on the two previous iterates, so it is not of the form $x_{k+1} = G(x_k)$ and the stable manifold argument does not apply directly. It is recovered by writing the iteration as a map on the pair (x_k, x_{k-1}) , at the cost of a non-symmetric differential.

Momentum can also be reset during the run. Restarting a momentum method means discarding the accumulated displacement and setting the momentum term to zero, so that the next step is a plain gradient step; the question is when to do it. Li and Lin [38] restart once the accumulated displacement exceeds a threshold, and obtain an ε -FOSP in $\mathcal{O}(\varepsilon^{-7/4})$ gradient evaluations for both the HB and accelerated variants. Perturbing the restart point when the gradient there is small strengthens the guarantee of the accelerated variant: it then returns an $(\varepsilon_g, \varepsilon_H)$ -SOSP in $\tilde{\mathcal{O}}(\varepsilon_g^{-7/4})$ gradient evaluations with high probability. This matches the best known rate for a perturbed first-order method.

2.6 Nonlinear conjugate gradient

The linear conjugate gradient method was introduced in 1952 for solving $Ax = b$ with A symmetric positive definite [46]. Writing $r_k := Ax_k - b$ for the residual, it generates

$$x_{k+1} = x_k + \alpha_k d_k, \quad d_{k+1} = -r_{k+1} + \beta_{k+1} d_k, \quad (2.1)$$

with $d_0 = -r_0$, the stepsize $\alpha_k = -r_k^\top d_k / (d_k^\top A d_k)$ minimizing the quadratic exactly along d_k , and $\beta_{k+1} = \|r_{k+1}\|^2 / \|r_k\|^2$ chosen so that the directions satisfy $d_i^\top A d_j = 0$ for $i \neq j$. Its extension to general smooth objectives [15] replaces the exact stepsize by a line search and the residual by the gradient, giving the two-term recurrence (1.6).

The choice of the coefficient β_{k+1} gives rise to the various NCG schemes [23, 54]. The Fletcher–Reeves (FR) formula [15],

$$\beta_{k+1}^{\text{FR}} := \frac{\|g_{k+1}\|^2}{\|g_k\|^2}, \quad (2.2)$$

converges globally under strong Wolfe conditions, but is prone to generating poor search directions and long sequences of tiny steps. The Polak–Ribière–Polyak (PRP) formula [52],

$$\beta_{k+1}^{\text{PRP}} := \frac{g_{k+1}^\top (g_{k+1} - g_k)}{\|g_k\|^2}, \quad (2.3)$$

behaves better in practice but can fail to converge. Truncating it at zero yields the PRP+ variant,

$$\beta_{k+1}^{\text{PRP}+} := \max\{\beta_{k+1}^{\text{PRP}}, 0\}, \quad (2.4)$$

which restores global convergence under appropriate line-search conditions [18]. Global convergence can also be recovered by keeping the PRP formula and modifying the line search instead [21, 22]. More recently, the formula of Hager and Zhang [24, 25],

$$\beta_{k+1}^{\text{HZ}} := \frac{1}{d_k^\top y_k} \left(y_k - 2d_k \frac{\|y_k\|^2}{d_k^\top y_k} \right)^\top g_{k+1}, \quad y_k := g_{k+1} - g_k, \quad (2.5)$$

has been found successful in practice.

The convergence results above are asymptotic. Some complexity bounds are known in specific convex cases. On quadratics, conjugate gradient is optimal while on general smooth convex functions it is not: [12] give the first non-asymptotic bounds for FR and PRP, together with examples on which neither improves on gradient descent with exact line search, while accelerated gradient is optimal up to constant factors. Karimi and Vavasis [31, 32, 33] study this gap in a series of works relating conjugate gradient to accelerated gradient. In particular, they propose a method that takes conjugate gradient steps until progress stalls, switches to accelerated gradient steps, and later retries conjugate gradient. It is identical to linear conjugate gradient on quadratics and attains the optimal rate for smooth convex functions [33].

In the nonconvex setting, complexity results are more recent. Whatever the variant, a NCG method can produce iterates for which $g_{k+1}^\top d_{k+1} \geq 0$, so that no decrease along d_{k+1} is guaranteed. The usual remedy is to discard the direction and set $d_{k+1} = -g_{k+1}$, a process known as restarting and a common feature of NCG implementations. Chan–Renous–Legoubin and Royer [7] propose an alternative restart condition that yields complexity guarantees: given $\sigma \in (0, 1]$, $\kappa \geq 1$, $p \geq 0$ and $q \geq 0$, the direction d_{k+1} is kept only if

$$g_{k+1}^\top d_{k+1} \leq -\sigma \|g_{k+1}\|^{1+p} \quad \text{and} \quad \|d_{k+1}\| \leq \kappa \|g_{k+1}\|^q, \quad (2.6)$$

and the method restarts otherwise. The first condition requires the direction to be sufficiently downhill, the second prevents its norm from growing too large relative to the gradient's. By partitioning the iterations according to whether (2.6) holds, they obtain the first worst-case complexity for reaching an ε_g -FOSP with NCG in the nonconvex setting.

No complexity guarantee for reaching an $(\varepsilon_g, \varepsilon_H)$ -SOSP is available for any NCG method, and neither of the two routes surveyed above applies. The dynamical-systems argument of Section 2.3 requires the maps G_k to be local diffeomorphisms of a common form, which the NCG recursion does not provide: it changes form at every restart, and its coefficient β_{k+1} depends nonlinearly on the current and previous gradients. Muşat and Boumal [40] encounter similar issues with the stepsize changing at each iteration due to the linesearch. As discussed in Section 1.1, this prevents the perturbation argument of Section 2.4 to apply directly to NCG. In the next chapter, however, we combine NCG with HB in order to apply the perturbation technique.

Chapter 3

Nonlinear Conjugate Gradient and Perturbed Heavy Ball

3.1 Setting, Algorithm, and Main Result

Throughout this work, we assume that the three following assumptions hold.

Assumption 1. $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and its gradient is L_1 -Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \leq L_1\|x - y\|$ for all x, y .

Assumption 2. $\nabla^2 f$ is L_2 -Lipschitz: $\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L_2\|x - y\|$ for all x, y .

Assumption 3. f is bounded below by f_{low} , and $\Delta_f := f(x_0) - f_{\text{low}} < \infty$.

The descent phase runs the safeguarded NCG method of [7], with the restart condition (2.6) and an Armijo backtracking line search. The escape phase, which is a perturbed version of the HB method from (1.4), is described below.

When $\|g_k\| \leq \varepsilon_g$ at some step k_{pert} , the iterate is a candidate SOSP. We freeze it as a snapshot $\tilde{x} := x_{k_{\text{pert}}}$ and draw $\xi \sim \text{Unif}(B_0(R))$, where R is the perturbation radius. From the perturbed point $\hat{x}_0 := \tilde{x} + \xi$, the escape phase runs T HB steps, indexed locally by t so that $\hat{x}_t = x_{k_{\text{pert}}+t}$,

$$\hat{x}_{t+1} = \hat{x}_t - \alpha_w g_t + \gamma(\hat{x}_t - \hat{x}_{t-1}), \quad t = 0, \dots, T-1, \quad (3.1)$$

with stepsize $\alpha_w = 1/L_1$, momentum coefficient $\gamma \in (0, \frac{1}{2})$ and $\hat{x}_{-1} := \hat{x}_0$.

At the end of the window we compare the resulting function value against the snapshot. We say the escape *succeeds* when

$$f(\tilde{x}) - f(\hat{x}_T) \geq F_{\text{cert}}, \quad (3.2)$$

that is, when the T HB steps decrease f by at least the certified amount F_{cert} . If (3.2) holds, $\tilde{x}$ is a strict saddle point, and the method discards it and returns to the descent phase. Otherwise $\tilde{x}$ is an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability, and is returned.

3.1.1 The algorithm

Each pass of the outer loop is either a descent step or a full escape window, according to whether the trigger $\|g_k\| \leq \varepsilon_g$ fires. A window consumes T inner iterations and returns either a certified SOSP, in which case the algorithm stops, or the window's last iterate $\hat{x}_T$, from which the descent phase resumes with a fresh direction $d = -g$. Accordingly, iterations are partitioned into

$$\begin{aligned} \mathcal{N} &= \{k \text{ descent step} : g_k^\top d_k \leq -\sigma \|g_k\|^{1+p} \text{ and } \|d_k\| \leq \kappa \|g_k\|^q\}, \\ \mathcal{R} &= \{\text{descent steps not in } \mathcal{N}\}, \quad \mathcal{W} = \{\text{escape-window steps}\}, \end{aligned}$$

Algorithm 2 PRNCG-HB: Perturbed restarted NCG with HB escape

```
1: Input:  $x_0 \in \mathbb{R}^n$ ;  $L_1, L_2$ ; tolerances  $\varepsilon_g, \varepsilon_H > 0$ ; Armijo  $\eta \in (0, 1)$ , backtracking  $\theta \in (0, 1)$ ;
   restart parameters  $\sigma \in (0, 1]$ ,  $\kappa \geq 1$ ,  $p \geq 0$ ,  $q \in [0, 1 + p]$ ; momentum  $\gamma \in (0, \frac{1}{2})$ ; failure
   probability  $\delta_{\text{tot}} \in (0, 1)$ ; gap bound  $\Delta_f$ ; escape window stepsize  $\alpha_w \leftarrow 1/L_1$ .
2: Derived constants (Section 3.4): window length  $T$ , perturbation radius  $R$ , target de-
   crease  $F$ , and slack  $\omega \in (0, 1)$ ; set certified decrease  $F_{\text{cert}} \leftarrow (1 - \omega)F$ .
3:  $g_0 \leftarrow \nabla f(x_0)$ ;  $d_0 \leftarrow -g_0$ 
4: for  $k = 0, 1, 2, \dots$  do
5:   if  $\|g_k\| \leq \varepsilon_g$  then ▷ candidate SOSP: perturb and run an escape window
6:      $\tilde{x} \leftarrow x_k$ 
7:      $\hat{x}_0 \leftarrow \tilde{x} + \xi$ ,  $\xi \sim \text{Unif}(B_0(R))$ 
8:      $\hat{x}_{-1} \leftarrow \hat{x}_0$  ▷ start from rest
9:     for  $t = 0, 1, \dots, T - 1$  do ▷ HB
10:       $\hat{x}_{t+1} \leftarrow \hat{x}_t - \alpha_w \nabla f(\hat{x}_t) + \gamma(\hat{x}_t - \hat{x}_{t-1})$ 
11:    end for
12:    if  $f(\tilde{x}) - f(\hat{x}_T) \geq F_{\text{cert}}$  then ▷ escape succeeded:  $\tilde{x}$  is a strict saddle (w.h.p.)
13:       $x_{k+1} \leftarrow \hat{x}_T$ 
14:       $g_{k+1} \leftarrow \nabla f(\hat{x}_T)$ 
15:       $d_{k+1} \leftarrow -g_{k+1}$  ▷ hand back to NCG, fresh direction
16:    else ▷ escape failed:  $\tilde{x}$  is an  $(\varepsilon_g, \varepsilon_H)$ -SOSP (w.h.p.)
17:      return  $\tilde{x}$  ▷ Theorem 1
18:    end if
19:  else ▷ descent phase: safeguarded NCG with backtracking
20:     $\alpha_k \leftarrow \theta^{j_k}$ ,  $j_k$  minimal with  $f(x_k + \theta^{j_k} d_k) \leq f(x_k) + \eta \theta^{j_k} g_k^\top d_k$ 
21:     $x_{k+1} \leftarrow x_k + \alpha_k d_k$ 
22:     $g_{k+1} \leftarrow \nabla f(x_{k+1})$ 
23:     $d_{k+1}^{\text{trial}} \leftarrow -g_{k+1} + \beta_{k+1} d_k$ 
24:    if (2.6) holds for  $d_{k+1}^{\text{trial}}$  then
25:       $d_{k+1} \leftarrow d_{k+1}^{\text{trial}}$ 
26:    else
27:       $d_{k+1} \leftarrow -g_{k+1}$ 
28:    end if
29:  end if
30: end for
```

where $\mathcal{W}$ collects the inner iterations of all windows, so that the total iteration count is $|\mathcal{N}| + |\mathcal{R}| + |\mathcal{W}|$. This partition drives the complexity analysis: $\mathcal{N}$ - and $\mathcal{R}$ -steps each yield a per-step decrease of f , with distinct constants $c_{\mathcal{N}}$ and $c_{\mathcal{R}}$ (Section 3.2), while $\mathcal{W}$ -steps are accounted for through the escape guarantee (Sections 3.3 and 3.4). The three rates of Theorem 1 correspond exactly to these three sets.

3.1.2 Main result

Theorem 1. *Let Assumptions 1 to 3 hold, let $\varepsilon_g \in (0, 1]$, $\varepsilon_H \in (0, L_1]$ and $\delta_{\text{tot}} \in (0, 1)$, and let the restart parameters satisfy $q \in [0, 1 + p]$. Run Algorithm 2 with the parameters of Proposition 1 and the failure budget (3.55), and let ϕ_0 be the logarithmic factor defined in Section 3.4.3. Then Algorithm 2 terminates and returns a point $\tilde{x}$ which, with probability at least $1 - \delta_{\text{tot}}$, is an*

$(\varepsilon_g, \varepsilon_H)$ -SOSP. The total number of iterations satisfies

$$N \leq \underbrace{\frac{\Delta f}{c_{\mathcal{R}}} \varepsilon_g^{-2} + \frac{\Delta f}{c_{\mathcal{N}}} \varepsilon_g^{-\max\{1+p, 2(1+p-q)\}}}_{\text{descent phase (Section 3.2)}} + \underbrace{C \frac{L_1 L_2^2 \Delta f \phi_0^4}{\varepsilon_H^4}}_{\text{escape phase (Sections 3.3 and 3.4)}},$$

where $c_{\mathcal{N}}, c_{\mathcal{R}}$ are the line-search constants of Lemmas 1 and 2 and C depends only on the momentum γ and the analysis parameters c, ω . Since ϕ_0 is logarithmic in the problem data, the escape term is $\tilde{\mathcal{O}}(\varepsilon_H^{-4})$.

Remark 1. Under the coupling $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$ used in [29, 30], the escape rate becomes $\tilde{\mathcal{O}}(\varepsilon_g^{-2})$, matching perturbed gradient descent [29].

Remark 2. Each escape-window iteration performs one gradient evaluation, plus one function evaluation per window for the certification test. A descent iteration performs one gradient evaluation and at most $\lfloor \bar{j}_{\mathcal{N},k} + 1 \rfloor$ (resp. $\bar{j}_{\mathcal{R}} + 1$) function evaluations for the backtracking line search (Lemmas 1 and 2).

3.2 The Descent Phase: Restarted NCG with Line Search

We recall the two decrease lemmas of [7] and derive a feature of the restart test used in Chapter 5. The lemmas supply the per-step decrease and the constants $c_{\mathcal{N}}, c_{\mathcal{R}}$ from the result of Theorem 1.

Lemma 1 ([7, Lemma 3.1]). *Let Assumption 1 hold and $k \in \mathcal{N}$ with $\|g_k\| > 0$. The backtracking terminates after at most $\lfloor \bar{j}_{\mathcal{N},k} + 1 \rfloor$ trials, with*

$$\bar{j}_{\mathcal{N},k} := \left\lceil \log_{\theta} \left(\frac{2(1-\eta)\sigma}{\kappa^2 L_1} \|g_k\|^{1+p-2q} \right) \right\rceil_+,$$

and

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{N}} \min \{ \|g_k\|^{1+p}, \|g_k\|^{2(1+p-q)} \}, \quad (3.3)$$

where

$$c_{\mathcal{N}} := \eta \sigma \min \left\{ 1, \frac{2(1-\eta)\sigma\theta}{\kappa^2 L_1} \right\}.$$

Lemma 2 ([7, Lemma 3.2]). *Let Assumption 1 hold and $k \in \mathcal{R}$ with $\|g_k\| > 0$. The backtracking terminates after at most $\bar{j}_{\mathcal{R}} + 1$ trials, with*

$$\bar{j}_{\mathcal{R}} := \left\lceil \log_{\theta} \left(\frac{2(1-\eta)}{L_1} \right) \right\rceil_+,$$

and

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{R}} \|g_k\|^2, \quad (3.4)$$

where

$$c_{\mathcal{R}} := \eta \min \left\{ 1, \frac{2(1-\eta)\theta}{L_1} \right\}.$$

Both lemmas give $f(x_{k+1}) < f(x_k)$: f is strictly decreasing along descent steps. These per-step decreases drive the descent-phase count in Section 3.4.

Lemma 3. *Let $k \in \mathcal{N}$ with $\|g_k\| > 0$. Then the search direction is sandwiched,*

$$\sigma \|g_k\|^p \leq \|d_k\| \leq \kappa \|g_k\|^q, \quad (3.5)$$

and consequently the safeguard forces the scale inequality

$$\sigma \|g_k\|^{p-q} \leq \kappa. \quad (3.6)$$

When $q > p$ the latter imposes a gradient floor,

$$\|g_k\| \geq (\sigma/\kappa)^{1/(q-p)}, \quad (3.7)$$

when $q < p$ it imposes a ceiling $\|g_k\| \leq (\kappa/\sigma)^{1/(p-q)}$, and when $q = p$ it imposes no constraint on the gradient scale.

Proof. For $k \in \mathcal{N}$ the restart test (2.6) holds, so $g_k^\top d_k \leq -\sigma \|g_k\|^{1+p}$ and hence $|g_k^\top d_k| \geq \sigma \|g_k\|^{1+p}$. Combining this with Cauchy–Schwarz and the norm bound in (2.6),

$$\sigma \|g_k\|^{1+p} \leq |g_k^\top d_k| \leq \|g_k\| \|d_k\| \leq \kappa \|g_k\|^{1+q}.$$

Dividing the chain by $\|g_k\|^{1+p} > 0$ gives the lower bound in (3.5), whose upper bound is the second condition of (2.6); dividing instead by $\|g_k\|^{1+q} > 0$ gives the inequality (3.6). Isolating $\|g_k\|$ according to the sign of $q-p$ then gives the floor (3.7) when $q > p$, the ceiling when $q < p$, and no constraint when $q = p$. $\square$

Remark 3. Geometrically, (3.5) confines $\|d_k\|$ to the band $[\sigma \|g_k\|^p, \kappa \|g_k\|^q]$, which is empty below the floor (3.7) when $q > p$: there the method restarts by necessity and degrades to GD. Remark 8 in Chapter 5.

3.3 The Escape Phase: Perturbed Heavy-Ball Windows

Following the perturbation strategy of [29, 30, 39], we show that when the snapshot has curvature below $-\varepsilon_H$, the perturbed HB window escapes it with high probability. The set of perturbations from which the window fails to escape forms a thin slab of the perturbation ball, which a uniformly drawn perturbation misses with high probability.

The proof runs in three steps, spread across the subsections as follows.

Step 1 (geometric separation; Sections 3.3.2 to 3.3.4). We run two copies of the escape phase from perturbations that differ by a small displacement along the direction of most negative curvature at the snapshot. We show that the gap between the two trajectories grows geometrically in the number of escape steps, at a rate $\mu_+ > 1$ set by that curvature.

Step 2 (incompatibility with localization; Section 3.3.1 and Section 3.3.4). If both runs of Step 1 fail to decrease f , each stays in a small ball around its own start, so the two trajectories remain within a bounded distance of one another. This contradicts the geometric growth of Step 1 once the window is long enough, and hence at most one of the two runs can fail.

Step 3 (volume argument; Section 3.3.5). Step 2 bounds how far apart two failing perturbations can be along the escape direction, so the set of such perturbations is thin in that direction. A volume estimate then shows that a uniformly random perturbation lands in it with low probability, which yields the escape guarantee.

3.3.1 Energy decrease and localization

Throughout this subsection we work inside a single escape window and use the associated notations: $\hat{x}_0, \dots, \hat{x}_T$ are the iterates of (3.1), $s_t := \hat{x}_{t+1} - \hat{x}_t$, and $\hat{x}_{-1} := \hat{x}_0$ gives $s_{-1} = 0$. Inside the window f does not need to decrease; the monotone quantity we track is the classical HB energy [65, 47]

$$E_t := f(\hat{x}_t) + \frac{\gamma}{2\alpha_w} \|s_{t-1}\|^2. \quad (3.8)$$

Lemma 4. *Let Assumptions 1 and 3 hold, $\gamma \in (0, \frac{1}{2})$, and $\alpha_w = 1/L_1$. Then for every window step $t \in \{0, \dots, T-1\}$,*

$$E_{t+1} \leq E_t - L_1 \left(\frac{1}{2} - \gamma \right) \|s_t\|^2. \quad (3.9)$$

Moreover $E_0 = f(\hat{x}_0)$ and $E_t \geq f(\hat{x}_t) \geq f_{\text{low}}$ for every t .

Proof. By L_1 -smoothness,

$$f(\hat{x}_{t+1}) \leq f(\hat{x}_t) + \langle g_t, s_t \rangle + \frac{L_1}{2} \|s_t\|^2.$$

By (3.1), $s_t = -\alpha_w g_t + \gamma s_{t-1}$, so $g_t = \frac{1}{\alpha_w}(\gamma s_{t-1} - s_t)$ and the linear term is

$$\langle g_t, s_t \rangle = \frac{\gamma}{\alpha_w} \langle s_{t-1}, s_t \rangle - \frac{1}{\alpha_w} \|s_t\|^2.$$

Bounding $\langle s_{t-1}, s_t \rangle \leq \frac{1}{2} \|s_{t-1}\|^2 + \frac{1}{2} \|s_t\|^2$ by Young's inequality yields

$$f(\hat{x}_{t+1}) \leq f(\hat{x}_t) + \frac{\gamma}{2\alpha_w} \|s_{t-1}\|^2 - \left(\frac{1}{\alpha_w} - \frac{\gamma}{2\alpha_w} - \frac{L_1}{2} \right) \|s_t\|^2.$$

Adding $\frac{\gamma}{2\alpha_w} \|s_t\|^2$ to both sides and using $\alpha_w = 1/L_1$ gives (3.9). Finally $s_{-1} = 0$ gives $E_0 = f(\hat{x}_0)$, and $E_t \geq f(\hat{x}_t) \geq f_{\text{low}}$ is immediate from (3.8) and Assumption 3. $\square$

Remark 4. Energies of the form $f(x) + \nu \|x - y\|^2$ are the standard Lyapunov functions for nonconvex HB [47, 65], and (3.8) is the case $\nu = \gamma/2\alpha_w$. The threshold $\gamma < \frac{1}{2}$ cannot be improved by choosing a different ν . Repeating the proof with a general coefficient $E_t^{(c)} = f(\hat{x}_t) + \frac{c}{2\alpha_w} \|s_{t-1}\|^2$ and the weighted Young inequality $\langle s_{t-1}, s_t \rangle \leq \frac{1}{2\tau} \|s_{t-1}\|^2 + \frac{\tau}{2} \|s_t\|^2$, a telescoping decrease requires $c \geq \gamma/\tau$ and $c < 1 - \gamma\tau$, and these are compatible for some $\tau > 0$ only when $\gamma < \frac{1}{2}$. Taking $\tau = 1$ and $c = \gamma$ recovers (3.8).

Lemma 5 (Improve-or-localize). *Under the setting of Lemma 4, for every $0 \leq t \leq T$,*

$$\|\hat{x}_t - \hat{x}_0\| \leq \sqrt{\frac{T(f(\hat{x}_0) - f(\hat{x}_T))}{L_1(\frac{1}{2} - \gamma)}}. \quad (3.10)$$

In particular $f(\hat{x}_0) - f(\hat{x}_T) \leq F$ implies $\|\hat{x}_t - \hat{x}_0\| \leq S$ for all $t \leq T$, where

$$S := \sqrt{\frac{TF}{L_1(\frac{1}{2} - \gamma)}}. \quad (3.11)$$

Proof. Telescoping (3.9) and using monotonicity of E ($E_t \geq E_T$) together with $E_0 = f(\hat{x}_0)$,

$$\begin{aligned} L_1 \left(\frac{1}{2} - \gamma \right) \sum_{j=0}^{t-1} \|s_j\|^2 &\leq E_0 - E_t \\ &\leq E_0 - E_T \\ &= f(\hat{x}_0) - f(\hat{x}_T) - \frac{\gamma}{2\alpha_w} \|s_{T-1}\|^2 \\ &\leq f(\hat{x}_0) - f(\hat{x}_T). \end{aligned}$$

By the triangle inequality and Cauchy-Schwarz, $\|\hat{x}_t - \hat{x}_0\| \leq \sqrt{t} (\sum_{j<t} \|s_j\|^2)^{1/2}$, and $t \leq T$ gives (3.10). $\square$

Lemma 5 is a dichotomy: if the window fails to decrease f by more than F , i.e. $f(\hat{x}_0) - f(\hat{x}_T) \leq F$, then every iterate stays within distance S of the start (*localize*). Equivalently, leaving the ball of radius S forces a function decrease exceeding F (*improve*). The escape analysis exploits the first statement: a non-escaping trajectory is confined to $B_{\hat{x}_0}(S)$.

3.3.2 The coupled difference recursion

We now set up the two-trajectory scheme announced in Step 1. Fix a snapshot $\tilde{x}$ at which a perturbation is added, write $H := \nabla^2 f(\tilde{x})$, and let $v_{\min}$ be a unit eigenvector associated with $\lambda_{\min}(H)$ (the escape direction). It is convenient to write the window (3.1) as:

$$d_t = -g_t + \gamma d_{t-1}, \quad \hat{x}_{t+1} = \hat{x}_t + \alpha_w d_t, \quad d_{-1} = 0, \quad (3.12)$$

which is equivalent to (3.1) since $\alpha_w d_t = \hat{x}_{t+1} - \hat{x}_t$.

Consider two runs of the same window, started from perturbations ξ and ξ' that differ by $\ell_0 > 0$ along $v_{\min}$:

$$\hat{x}_0 = \tilde{x} + \xi, \quad \hat{x}'_0 = \tilde{x} + \xi', \quad \xi' = \xi + \ell_0 v_{\min}, \quad \xi, \xi' \in B_0(R),$$

and set $\Delta x_t := \hat{x}_t - \hat{x}'_t$ and $\Delta d_t := d_t - d'_t$. The rescaled difference state is

$$w_t := \begin{pmatrix} \Delta x_t \\ \alpha_w \Delta d_{t-1} \end{pmatrix} \in \mathbb{R}^{2n}, \quad w_0 = \begin{pmatrix} -\ell_0 v_{\min} \\ 0 \end{pmatrix}, \quad \|w_0\| = \ell_0. \quad (3.13)$$

Note that w_0 is aligned with the escape direction: its first component is a multiple of $v_{\min}$ and its second vanishes. This alignment places w_0 in the two-dimensional invariant plane of the block associated with $\lambda_{\min}$, and is used in Lemma 11. Finally, write the gradient as its linear part at the snapshot plus a remainder, $g_t = H\hat{x}_t + e_t$ with $e_t := \nabla f(\hat{x}_t) - H\hat{x}_t$.

Lemma 6. *Along an escape window, with constant stepsize $\alpha_w = 1/L_1$ and constant momentum γ , the difference state obeys*

$$w_{t+1} = \mathbf{M} w_t + u_t, \quad \mathbf{M} = \begin{pmatrix} I - \alpha_w H & \gamma I \\ -\alpha_w H & \gamma I \end{pmatrix}, \quad u_t = \begin{pmatrix} -\alpha_w \Delta e_t \\ -\alpha_w \Delta e_t \end{pmatrix}, \quad (3.14)$$

where $\Delta e_t = \nabla f(\hat{x}_t) - \nabla f(\hat{x}'_t) - H\Delta x_t$. If moreover both runs remain localized in the sense of Lemma 5, that is

$$\|\hat{x}_t - \tilde{x}\| \leq S \quad \text{and} \quad \|\hat{x}'_t - \tilde{x}\| \leq S \quad \text{for all } 0 \leq t \leq T,$$

then the remainder is proportional to the gap:

$$\|u_t\| \leq \chi \|w_t\|, \quad \chi := \frac{\sqrt{2} L_2 (S + R)}{L_1}. \quad (3.15)$$

Proof. From $\hat{x}_{t+1} = \hat{x}_t + \alpha_w d_t$ and $d_t = -(H\hat{x}_t + e_t) + \gamma d_{t-1}$, differencing the two runs gives

$$\alpha_w \Delta d_t = -\alpha_w H \Delta x_t - \alpha_w \Delta e_t + \gamma (\alpha_w \Delta d_{t-1}),$$

$$\Delta x_{t+1} = \Delta x_t + \alpha_w \Delta d_t = (I - \alpha_w H) \Delta x_t + \gamma (\alpha_w \Delta d_{t-1}) - \alpha_w \Delta e_t,$$

which is exactly (3.14). The stepsize α_w and the momentum γ are the *same* in both runs, so they factor out of the differences; this is what makes $\mathbf{M}$ constant and independent of t .

For the remainder bound, apply the fundamental theorem of calculus to ∇f along the segment joining $\hat{x}'_t$ to $\hat{x}_t$:

$$\Delta e_t = \left(\int_0^1 (\nabla^2 f(\hat{x}'_t + r \Delta x_t) - H) dr \right) \Delta x_t,$$

and by Assumption 2, $\|\nabla^2 f(\hat{x}'_t + r \Delta x_t) - H\| \leq L_2 \|\hat{x}'_t + r \Delta x_t - \tilde{x}\|$ for every $r \in [0, 1]$. The map $r \mapsto \|\hat{x}'_t + r \Delta x_t - \tilde{x}\|$ is convex, so it attains its maximum over $[0, 1]$ at an endpoint, that is at $\hat{x}_t$ or $\hat{x}'_t$. Both endpoints are bounded by the localization hypothesis and $\xi \in B_0(R)$:

$$\|\hat{x}_t - \tilde{x}\| \leq \|\hat{x}_t - \hat{x}_0\| + \|\hat{x}_0 - \tilde{x}\| \leq S + R,$$

and identically for the primed run. Hence

$$\|\Delta e_t\| \leq \left(\int_0^1 L_2 \|\hat{x}'_t + r \Delta x_t - \tilde{x}\| dr \right) \|\Delta x_t\| \leq L_2 (S + R) \|\Delta x_t\|.$$

Finally, both components of u_t equal $-\alpha_w \Delta e_t$, so $\|u_t\| = \sqrt{2} \alpha_w \|\Delta e_t\|$, and Δx_t is the first block of w_t , so $\|\Delta x_t\| \leq \|w_t\|$. Combining with $\alpha_w = 1/L_1$ gives (3.15). $\square$

Remark 5. Section 1.1 stated that the proof technique requires both runs to use the same stepsize and momentum coefficient at every step. (3.14) shows why it is required: *the remainder must be proportional to the gap*. Suppose the two runs used per-iteration stepsizes α_t and α'_t . Differencing $\hat{x}_{t+1} = \hat{x}_t + \alpha_t d_t$ and $\hat{x}'_{t+1} = \hat{x}'_t + \alpha'_t d'_t$ gives

$$\Delta x_{t+1} = \Delta x_t + \alpha_t \Delta d_t + (\alpha_t - \alpha'_t) d'_t,$$

and the last term involves the *full* search direction d'_t . Its magnitude is of order $\|d'_t\|$, whereas every other term is of order $\|w_t\|$, that is, of order of the gap between the trajectories. Consequently the bound $\|u_t\| \leq \chi \|w_t\|$ of (3.15) fails, and with it the induction of Lemma 12, at its first step. A per-run momentum coefficient produces the same obstruction with the term $(\gamma_t - \gamma'_t) d'_{t-1}$, and the restart test is the extreme case: when the two runs make opposite restart decisions the coefficient jumps between β_t and 0, so $\beta_t - \beta'_t$ is as large as β_t itself and bears no relation to the gap.

M must be constant. Even with parameters shared between the two runs, letting them vary in t would replace **M** by a sequence $\mathbf{M}_t$, so that iterating (3.14) would produce the product $\mathbf{M}_{T-1} \cdots \mathbf{M}_0$ in place of the power $\mathbf{M}^T$. The analysis of Section 3.3.3 bounds powers of a *fixed* matrix through its eigenstructure (Lemmas 7 and 10); products of distinct matrices admit no such description, and neither the uniform rate μ_+ nor the two-sided estimate of Lemma 11 survives.

3.3.3 Spectral analysis of the matrix **M**

The recursion (3.14) is driven by the matrix **M** and perturbed by the remainder u_t . We need two bounds on **M**. First, $\mathbf{M}^t w_0$ must grow geometrically at some rate, which is what separates the two trajectories. Second, no power $\mathbf{M}^m$ may grow faster than that same rate, so that the accumulated remainder cannot overtake the separation. Both bounds must carry explicit constants, and both come from the eigenstructure of **M**.

For PGD the same argument runs with $I - \alpha_w H$ in place of **M**. That matrix is symmetric, so it diagonalizes in the eigenbasis of H and its eigenvalues give both bounds directly. Here w_t stacks a position difference and a direction difference, so **M** acts on $\mathbb{R}^{2n}$ and is not symmetric: it does not diagonalize in that basis. It does, however, split into 2×2 blocks along the eigenbasis of H , one block per eigenvalue, and the rest of the analysis works with those blocks.

Lemma 7. *Let $H = VDV^\top$ with V orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. In the reordered basis grouping the i -th coordinates of the two components of (3.13), **M** is orthogonally similar to $\text{blockdiag}(B_1, \dots, B_n)$ with*

$$B_i = \begin{pmatrix} 1 - \alpha_w \lambda_i & \gamma \\ -\alpha_w \lambda_i & \gamma \end{pmatrix}, \quad \det B_i = \gamma, \quad \text{tr } B_i = 1 - \alpha_w \lambda_i + \gamma. \quad (3.16)$$

In particular, since the similarity is orthogonal,

$$\|\mathbf{M}^m\| = \max_i \|B_i^m\| \quad \text{for every } m \geq 0. \quad (3.17)$$

Proof. Set $P := \text{diag}(V, V) \in \mathbb{R}^{2n \times 2n}$, which is orthogonal. Recall from (3.13) that $w = (\Delta x, \alpha_w \Delta d)$, so the first block of $2n$ coordinates carries the position difference and the second carries the rescaled direction difference. Conjugating **M** by P applies $V^\top(\cdot)V$ to each $n \times n$ sub-block, sending H to D and I to I ; every sub-block thereby becomes diagonal:

$$P^\top \mathbf{M} P = \begin{pmatrix} V^\top (I - \alpha_w H) V & \gamma V^\top V \\ -\alpha_w V^\top H V & \gamma V^\top V \end{pmatrix} = \begin{pmatrix} I - \alpha_w D & \gamma I \\ -\alpha_w D & \gamma I \end{pmatrix}. \quad (3.18)$$

In the new coordinates, write $\Delta x^{(i)}$ and $\alpha_w \Delta d^{(i)}$ for the components of $V^\top \Delta x$ and $\alpha_w V^\top \Delta d$ along the eigenvector v_i of H . Every sub-block of (3.18) being diagonal, the pair $(\Delta x^{(i)}, \alpha_w \Delta d^{(i)})$ evolves independently of all other indices: the only nonzero entries of $P^\top \mathbf{M} P$ couple these two coordinates to each other. Let Π be the permutation matrix reordering the $2n$ coordinates into the consecutive pairs $(i, n+i)$, $i = 1, \dots, n$. Then $P\Pi$ is orthogonal and gathers the coupled entries into n diagonal 2×2 blocks:

$$(P\Pi)^\top \mathbf{M} (P\Pi) = \text{blockdiag}(B_1, \dots, B_n), \quad B_i = \begin{pmatrix} 1 - \alpha_w \lambda_i & \gamma \\ -\alpha_w \lambda_i & \gamma \end{pmatrix}. \quad (3.19)$$

A direct computation gives $\det B_i = \gamma(1 - \alpha_w \lambda_i) + \gamma \alpha_w \lambda_i = \gamma$ and $\text{tr } B_i = 1 - \alpha_w \lambda_i + \gamma$, which is (3.16). Since the spectral norm is invariant under orthogonal similarity and the block-diagonal form has $\text{blockdiag}(B_1, \dots, B_n)^m = \text{blockdiag}(B_1^m, \dots, B_n^m)$, we obtain (3.17). $\square$

Throughout the spectral analysis we treat the blocks of Lemma 7 as a one-parameter family. Since $\|H\| \leq L_1$ by Assumption 1, every eigenvalue of H lies in $[\lambda_{\min}, L_1]$, and for λ in that range we set

$$B(\lambda) = \begin{pmatrix} 1 - \alpha_w \lambda & \gamma \\ -\alpha_w \lambda & \gamma \end{pmatrix},$$

so that, by Lemma 7, $B(\lambda)$ has constant determinant and λ -dependent trace

$$\det B(\lambda) = \gamma, \quad t(\lambda) := \text{tr } B(\lambda) = 1 - \alpha_w \lambda + \gamma,$$

characteristic polynomial $\mu^2 - t(\lambda)\mu + \gamma$, and discriminant

$$\Delta(\lambda) := t(\lambda)^2 - 4\gamma.$$

When $\Delta(\lambda) \geq 0$ the eigenvalues are real, $\mu_\pm(\lambda) = \frac{1}{2}(t(\lambda) \pm \sqrt{\Delta(\lambda)})$. When $\Delta(\lambda) < 0$ they are complex conjugates of modulus $\sqrt{\det B(\lambda)} = \sqrt{\gamma}$. We write $\rho(\lambda)$ for the spectral radius of $B(\lambda)$, so that $|\mu_+(\lambda) - \mu_-(\lambda)| = \sqrt{|\Delta(\lambda)|}$ in the real case. The block driving the escape is $B_{\min} := B(\lambda_{\min})$, whose eigenvalues we abbreviate $\mu_+ := \mu_+(\lambda_{\min})$ and $\mu_- := \mu_-(\lambda_{\min})$; in particular $\mu_+ \mu_- = \gamma$ and $t(\lambda_{\min}) = 1 + \gamma + \alpha_w |\lambda_{\min}|$.

One elementary fact is used repeatedly: t is strictly decreasing in λ (since $\alpha_w > 0$) and remains positive on the whole range, because with $\alpha_w = 1/L_1$,

$$t(\lambda) \geq t(L_1) = 1 - \alpha_w L_1 + \gamma = \gamma > 0 \quad (\lambda \in [\lambda_{\min}, L_1]). \quad (3.20)$$

In the real regime both roots are therefore positive and $\rho(\lambda) = \mu_+(\lambda)$.

Bounding powers of these blocks at a prescribed geometric rate is a purely linear-algebraic matter, deferred to Chapter A. We use two bounds there, according to whether the block's eigenvalues are separated or nearly equal. The first, Lemma 24, applies to any 2×2 matrix B with distinct eigenvalues $\mu_1 \neq \mu_2$ and spectral radius ρ , and reads

$$\|B^m\| \leq C(B) \rho^m, \quad C(B) := \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|}, \quad (3.21)$$

so its constant blows up as the eigenvalue gap closes. The second, Lemma 25, bounds $\|B^m\|$ through a Schur form and never divides by the gap, at the price of a factor m ; it is stated in (A.3).

The block associated with the most negative curvature drives the escape, and must have a real eigenvalue strictly larger than one for geometric growth to be possible.

Lemma 8. Let $\lambda_{\min} \leq -\varepsilon_H$ and let $p(\mu) = \mu^2 - t\mu + \gamma$ be the characteristic polynomial of $B_{\min}$, with $t = 1 + \gamma + \alpha_w |\lambda_{\min}|$. Then $B_{\min}$ has real eigenvalues $\mu_- < 1 < \mu_+$, and

$$\mu_+ \geq 1 + \alpha_w |\lambda_{\min}| \geq 1 + \alpha_w \varepsilon_H. \quad (3.22)$$

Proof. Write $a := \alpha_w |\lambda_{\min}| > 0$, so that $t = 1 + \gamma + a$. Evaluating at 1,

$$p(1) = 1 - t + \gamma = -a < 0.$$

Since p is an upward parabola, a strictly negative value at 1 places 1 strictly between its two roots; in particular both roots are real and $\mu_- < 1 < \mu_+$. Evaluating at $1 + a$,

$$p(1 + a) = (1 + a)[(1 + a) - t] + \gamma = (1 + a)(-\gamma) + \gamma = -\gamma a \leq 0,$$

so $1 + a$ also lies between the roots, and $\mu_+ \geq 1 + a = 1 + \alpha_w |\lambda_{\min}|$. The second inequality in (3.22) follows from $|\lambda_{\min}| \geq \varepsilon_H$. $\square$

By Lemma 7 the spectral radius of $\mathbf{M}$ is $\max_i \rho(B_i)$. The next lemma shows this maximum is attained at the escape block, so that $\rho(\mathbf{M}) = \mu_+$.

Lemma 9. For every $\lambda \in [\lambda_{\min}, L_1]$, the spectral radius of $B(\lambda)$ satisfies $\rho(\lambda) \leq \mu_+$.

Proof. If $\Delta(\lambda) < 0$, the eigenvalues are complex conjugates of modulus $\rho(\lambda) = \sqrt{\gamma} < 1 < \mu_+$, using $\gamma < \frac{1}{2}$ and Lemma 8.

If $\Delta(\lambda) \geq 0$, the eigenvalues are real and, by (3.20), positive; moreover $t(\lambda)^2 \geq 4\gamma$ and $t(\lambda) > 0$ give $t(\lambda) \geq 2\sqrt{\gamma}$, so

$$\rho(\lambda) = \mu_+(\lambda) = \frac{1}{2}(t(\lambda) + \sqrt{t(\lambda)^2 - 4\gamma}).$$

On $\{t \geq 2\sqrt{\gamma}\}$ the map $t \mapsto \frac{1}{2}(t + \sqrt{t^2 - 4\gamma})$ is increasing, being a sum of increasing functions of t . Since t is strictly decreasing in λ , it attains its maximum over $[\lambda_{\min}, L_1]$ at $\lambda_{\min}$, where $t(\lambda_{\min}) \geq 1 + \gamma \geq 2\sqrt{\gamma}$ by the AM–GM inequality. Hence $\mu_+(\lambda) \leq \mu_+(\lambda_{\min}) = \mu_+$. $\square$

Lemmas 9 and 24 together give $\|\mathbf{M}^m\| \leq \max_i C(B_i) \mu_+^m$, but the constant is not bounded over all Hessians. Indeed the constant $C(B)$ of (3.21) divides by the gap $|\mu_1 - \mu_2|$ between the two eigenvalues, and that gap vanishes when $\Delta(\lambda) = 0$, that is at $\lambda^* = L_1(1 - \sqrt{\gamma})^2$, where both eigenvalues equal $\sqrt{\gamma}$.

At λ^* the obstruction is not that the bound of Lemma 24 is loose but that no bound at rate ρ exists at all. Since its off-diagonal entry γ is nonzero, $B(\lambda^*)$ is not scalar and is therefore defective, hence similar to a Jordan block, whose powers

$$\begin{pmatrix} \mu & 1 \\ 0 & \mu \end{pmatrix}^m = \begin{pmatrix} \mu^m & m\mu^{m-1} \\ 0 & \mu^m \end{pmatrix}$$

carry a factor $m\mu^{m-1}$ off the diagonal. Consequently $\|B(\lambda^*)^m\|$ is of order $m\rho^{m-1}$ with $\rho = \sqrt{\gamma}$, so $\|B(\lambda^*)^m\|/\rho^m \rightarrow \infty$ and no constant C can satisfy $\|B(\lambda^*)^m\| \leq C\rho^m$ for all m .

The situation is saved by the location of the degeneracy. It occurs only at $\lambda^* > 0$, where the spectral radius $\sqrt{\gamma}$ is bounded below one, whereas the target rate satisfies $\mu_+ > 1$. Bounding every block at the common rate μ_+ rather than at its own ρ absorbs the polynomial factor, since $m\rho^{m-1} \leq C(\gamma) \mu_+^m$ whenever $\rho < 1 < \mu_+$. Lemma 10 makes this precise, with a constant depending on γ and no division by the eigenvalue gap.

Lemma 10. Let $\gamma \in (0, 1)$ (in particular any $\gamma < \frac{1}{2}$ as used in Algorithm 2) and $\lambda_{\min} < 0$. Set

$$\varsigma := \frac{1 - \gamma}{2}, \quad \bar{\rho} := \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2}, \quad b_{\mathbf{M}} := \max_i \|B_i\|_F.$$

Then $\bar{\rho} < 1$, $b_{\mathbf{M}} \leq \sqrt{5 + 2\gamma^2}$, and for every $m \geq 0$,

$$\|\mathbf{M}^m\| \leq C_{\mathbf{M}} \mu_+^m, \quad C_{\mathbf{M}} := \max \left\{ \frac{4(b_{\mathbf{M}} + \mu_+)}{1 - \gamma}, \sqrt{2} + \frac{b_{\mathbf{M}}}{(1 - \bar{\rho})^2} \right\}. \quad (3.23)$$

Proof. By (3.17) it suffices to bound $\|B(\lambda)^m\| \leq C_{\mathbf{M}}\mu_+^m$ for each eigenvalue $\lambda \in [\lambda_{\min}, L_1]$.

The bound of Lemma 24 does not apply when the two eigenvalues are close, since its constant divides by $|\mu_1 - \mu_2| = |\Delta(\lambda)|^{1/2}$. We therefore split the spectrum at the threshold ς : blocks with gap at least ς are handled by Lemma 24, while blocks with smaller gap, which occur only at $\lambda > 0$ where $\rho < 1$, are handled by Lemma 25.

A bound on $b_{\mathbf{M}}$ comes naturally from the parameters: since $\alpha_w = 1/L_1$ and $|\lambda_i| \leq L_1$, we have $|\alpha_w \lambda_i| \leq 1$, so

$$\|B_i\|_F^2 = (1 - \alpha_w \lambda_i)^2 + (\alpha_w \lambda_i)^2 + 2\gamma^2 \leq 4 + 1 + 2\gamma^2 = 5 + 2\gamma^2.$$

Likewise, $\bar{\rho} < 1$ is immediate. Indeed, as $2 - \varsigma = \frac{3+\gamma}{2} > 0$, both sides of the defining inequality may be squared:

$$\begin{aligned} \bar{\rho} < 1 &\iff \sqrt{4\gamma + \varsigma^2} < 2 - \varsigma \\ &\iff 4\gamma < 4 - 4\varsigma \\ &\iff \gamma < 1 - \varsigma = \frac{1+\gamma}{2} \\ &\iff \gamma < 1, \end{aligned}$$

which holds by hypothesis.

We now proceed to identify that the degeneracy sits at $\lambda > 0$. If $\lambda \leq 0$ then $t(\lambda) \geq 1 + \gamma$, so

$$\Delta(\lambda) \geq (1 + \gamma)^2 - 4\gamma = (1 - \gamma)^2 = 4\varsigma^2 \geq \varsigma^2. \quad (3.24)$$

Every block with $\lambda \leq 0$, and in particular the escape block $B_{\min}$, therefore has gap at least ς , so the near-degenerate regime can occur only at $\lambda > 0$. The rest of the proof handles both cases.

Case 1 (large gap): $|\Delta(\lambda)| \geq \varsigma^2$. The eigenvalues are distinct, so Lemma 24 applies. With $\|B\| \leq \|B\|_F \leq b_{\mathbf{M}}$, the bound $\rho(\lambda) \leq \mu_+$ of Lemma 9, and $|\mu_1 - \mu_2| = \sqrt{|\Delta(\lambda)|} \geq \varsigma$,

$$\begin{aligned} \|B^m\| &\leq \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|} \rho^m \\ &\leq \frac{2(b_{\mathbf{M}} + \mu_+)}{\varsigma} \mu_+^m \\ &= \frac{4(b_{\mathbf{M}} + \mu_+)}{1 - \gamma} \mu_+^m. \end{aligned} \quad (3.25)$$

Case 2 (small gap): $|\Delta(\lambda)| < \varsigma^2$. By (3.24) this forces $\lambda > 0$. We first show that the spectral radius is bounded away from 1, then bound $\|B^m\|$ without dividing by the gap.

(a) *The spectral radius satisfies $\rho \leq \bar{\rho} < 1$.* Two cases, according to the sign of $\Delta(\lambda)$.

If $\Delta(\lambda) \leq 0$, the eigenvalues are complex conjugate of modulus $\rho = \sqrt{\gamma}$, and since $\varsigma^2 \geq 0$,

$$\bar{\rho} = \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2} \geq \frac{2\sqrt{\gamma}}{2} = \sqrt{\gamma} = \rho.$$

If $0 < \Delta(\lambda) < \varsigma^2$, the eigenvalues are real and positive by (3.20), so $t(\lambda) = \sqrt{\Delta(\lambda) + 4\gamma}$. Both terms of ρ are then bounded above:

$$t(\lambda) < \sqrt{\varsigma^2 + 4\gamma}, \quad \sqrt{\Delta(\lambda)} < \varsigma,$$

hence

$$\rho = \frac{t(\lambda) + \sqrt{\Delta(\lambda)}}{2} < \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2} = \bar{\rho}.$$

In both cases $\rho \leq \bar{\rho}$, and $\bar{\rho} < 1$ by Step 1.

(b) *Bounding $\|B^m\|$.* The gap may now be arbitrarily small, so we replace Lemma 24 by Lemma 25. With $\|B\|_F \leq b_{\mathbf{M}}$, (A.3) gives

$$\|B^m\| \leq \sqrt{2} \rho^m + b_{\mathbf{M}} m \rho^{m-1}.$$

It remains to absorb the factor m . Since $\rho \leq \bar{\rho} < 1$, the term $m \rho^{m-1}$ is one summand of the convergent series $\sum_{l \geq 1} l \bar{\rho}^{l-1} = (1 - \bar{\rho})^{-2}$, so

$$m \rho^{m-1} \leq \frac{1}{(1 - \bar{\rho})^2},$$

while $\rho^m \leq 1 \leq \mu_+^m$ because $\mu_+ > 1$. Combining the last three displays,

$$\|B^m\| \leq \left(\sqrt{2} + \frac{b_{\mathbf{M}}}{(1 - \bar{\rho})^2} \right) \mu_+^m. \quad (3.26)$$

Bounds (3.25) and (3.26) cover every block for all $m \geq 1$; for $m = 0$ each side is at least $\|I\| = 1 \leq C_{\mathbf{M}}$. Taking the larger of the two constants gives $\|B_i^m\| \leq C_{\mathbf{M}} \mu_+^m$ for every i , and (3.23) follows from (3.17). $\square$

It remains to show that the homogeneous part $h_t := \mathbf{M}^t w_0$ actually attains the rate μ_+ .

Lemma 11. *Let $h_t = \mathbf{M}^t w_0$ be the homogeneous part of the recursion (3.14), with w_0 as in (3.13). Then h_t remains in the two-dimensional invariant plane associated with $\lambda_{\min}$, where its coordinate pair evolves as $h_t^{\min} = B_{\min}^t w_0^{\min}$ with $w_0^{\min} = (-\ell_0, 0)$. Writing Δx_t^{hom} for the first coordinate of $h_t^{\min}$, we have for every $t \geq 0$*

$$\|h_t\| \geq |\Delta x_t^{\text{hom}}| \geq c_{\text{esc}} \mu_+^t \ell_0, \quad c_{\text{esc}} := \frac{\mu_+ - \gamma}{\mu_+ - \mu_-}, \quad (3.27)$$

and

$$\|h_t\| \leq c_h \mu_+^t \ell_0, \quad c_h := C(B_{\min}) \leq \frac{2(b_{\mathbf{M}} + \mu_+)}{1 - \gamma}, \quad (3.28)$$

where $C(\cdot)$ is as in (3.21).

Proof. Reduction to the escape block. By Lemma 7, the orthogonal change of basis $P\Pi$ turns $\mathbf{M}$ into blockdiag($B_1, \dots, B_n$), grouping the $2n$ coordinates into n pairs, with block i acting on $(\Delta x^{(i)}, \alpha_w \Delta d^{(i)})$.

In this basis, write w_0 as its n coordinate pairs. Its second component is $\alpha_w \Delta d_{-1} = 0$, so the second entry of every pair vanishes. Its first component is $\Delta x_0 = -\ell_0 v_{\min}$ by (3.13), which has a single nonzero coordinate in the eigenbasis of H , the one along $v_{\min}$. Hence every pair of w_0 vanishes except the one associated with $\lambda_{\min}$, which equals

$$w_0^{\min} = (-\ell_0, 0).$$

Since the block-diagonal form leaves each pair invariant, $h_t = \mathbf{M}^t w_0$ remains in that plane for all t , and its surviving pair evolves as

$$h_t^{\min} = B_{\min}^t w_0^{\min}.$$

Because $P\Pi$ is orthogonal, norms are preserved, so we may work in the plane throughout.

Eigen-decomposition of $w_0^{\min}$. By Lemma 8 the eigenvalues $\mu_- < 1 < \mu_+$ of $B_{\min}$ are real and distinct, so the associated eigenvectors form a basis of the plane. For $\mu \in \{\mu_+, \mu_-\}$, writing

$v = (v^{(1)}, v^{(2)})$ for an eigenvector of $B_{\min}$ associated with μ , the second row of $(B_{\min} - \mu I)v = 0$ reads

$$\alpha_w \lambda_{\min} v^{(1)} = (\gamma - \mu) v^{(2)},$$

so we may take the unnormalized eigenvectors

$$v_\mu = \begin{pmatrix} \gamma - \mu \\ \alpha_w \lambda_{\min} \end{pmatrix}, \quad \mu \in \{\mu_+, \mu_-\},$$

abbreviated $v_+ := v_{\mu_+}$ and $v_- := v_{\mu_-}$. They have the same second entry.

Let $\pi_+, \pi_- \in \mathbb{R}$ be the coordinates of $w_0^{\min}$ in this basis, so that $w_0^{\min} = \pi_+ v_+ + \pi_- v_-$. Reading the two coordinates of $w_0^{\min} = (-\ell_0, 0)$ in turn, the second gives $\alpha_w \lambda_{\min}(\pi_+ + \pi_-) = 0$, hence

$$\pi_- = -\pi_+,$$

and the first then gives

$$-\ell_0 = \pi_+(\gamma - \mu_+) + \pi_-(\gamma - \mu_-) = \pi_+[(\gamma - \mu_+) - (\gamma - \mu_-)] = \pi_+(\mu_- - \mu_+),$$

so that

$$|\pi_+| = \frac{\ell_0}{\mu_+ - \mu_-}. \quad (3.29)$$

Lower bound. Write $\zeta := \mu_-/\mu_+ \in (0, 1)$. Applying $B_{\min}^t$ to $w_0^{\min} = \pi_+(v_+ - v_-)$,

$$h_t^{\min} = \pi_+ \mu_+^t (v_+ - \zeta^t v_-),$$

whose first coordinate is

$$\Delta x_t^{\text{hom}} = \pi_+ \mu_+^t [(\gamma - \mu_+) - \zeta^t(\gamma - \mu_-)].$$

We bound the bracket from below. Since $\mu_+ \mu_- = \gamma$ and $\mu_+ > 1 > \gamma$, we have $\mu_- = \gamma/\mu_+ < \gamma$, so

$$\gamma - \mu_+ < 0 \quad \text{and} \quad \gamma - \mu_- > 0.$$

Both terms of the bracket are therefore negative, and dropping the second one can only decrease its magnitude:

$$\left| (\gamma - \mu_+) - \zeta^t(\gamma - \mu_-) \right| \geq |\gamma - \mu_+| = \mu_+ - \gamma \quad \text{for every } t \geq 0.$$

Together with (3.29) this gives

$$|\Delta x_t^{\text{hom}}| \geq \frac{\mu_+ - \gamma}{\mu_+ - \mu_-} \mu_+^t \ell_0 = c_{\text{esc}} \mu_+^t \ell_0.$$

Finally Δx_t^{hom} is a coordinate of $h_t^{\min}$, so $\|h_t\| = \|h_t^{\min}\| \geq |\Delta x_t^{\text{hom}}|$, which completes (3.27).

Upper bound. We bound the constant $C(B_{\min})$ of (3.21). Since $\lambda_{\min} \leq 0$ we have $t(\lambda_{\min}) \geq 1 + \gamma$, so the eigenvalue gap satisfies

$$\mu_+ - \mu_- = \sqrt{\Delta(\lambda_{\min})} \geq \sqrt{(1 + \gamma)^2 - 4\gamma} = 1 - \gamma,$$

while $\|B_{\min}\| \leq \|B_{\min}\|_F \leq b_{\mathbf{M}}$. Hence

$$C(B_{\min}) = \frac{2(\|B_{\min}\| + \mu_+)}{\mu_+ - \mu_-} \leq \frac{2(b_{\mathbf{M}} + \mu_+)}{1 - \gamma}.$$

Lemma 24 with $\rho(B_{\min}) = \mu_+$ then gives

$$\|h_t\| \leq \|B_{\min}^t\| \|w_0^{\min}\| \leq C(B_{\min}) \mu_+^t \ell_0 = c_h \mu_+^t \ell_0.$$

□

This subsection delivers the two-sided estimate that drives Section 3.3.4: the homogeneous part of the difference grows at least like $c_{\text{esc}} \mu_+^t \ell_0$ for every t (Lemma 11), while powers of $\mathbf{M}$ are bounded by $C_{\mathbf{M}} \mu_+^m$ uniformly over the spectrum of H (Lemma 10). The same rate appears on both sides, which is what the induction of Lemma 12 requires.

3.3.4 Geometric growth of the gap

Lemmas 7 to 11 show that the homogeneous part of the gap grows geometrically. It remains to check that the accumulated remainder does not cancel that growth. Split the difference state as $w_t = h_t + z_t$, where

$$h_t = \mathbf{M}^t w_0, \quad z_t = \sum_{\tau=0}^{t-1} \mathbf{M}^{t-1-\tau} u_\tau,$$

so that h_t is the homogeneous solution of Lemma 11 and z_t collects the remainders of (3.14) propagated forward.

Lemma 12. *Consider an escape window in which both trajectories remain localized, that is $\|\hat{x}_t - \hat{x}_0\| \leq S$ and $\|\hat{x}'_t - \hat{x}'_0\| \leq S$ for all $t \leq T$, so that (3.15) applies. Let $c \in (0, \min\{1, c_{\text{esc}}/c_h\})$ and suppose*

$$\chi \leq \frac{c c_{\text{esc}} \mu_+}{C_{\mathbf{M}} c_h (1+c)(T+1)} =: \frac{\Lambda}{T+1}. \quad (3.30)$$

Then $\|z_t\| \leq c \|h_t\|$ for every $t \leq T$, and in particular

$$\|\Delta x_T\| \geq (c_{\text{esc}} - c c_h) \mu_+^T \ell_0. \quad (3.31)$$

Proof. We prove $\|z_t\| \leq c \|h_t\|$ for all $t \leq T$ by induction on t .

Base case. $z_0 = 0$, so the claim holds at $t = 0$.

Induction step. Let $0 \leq t \leq T-1$ and assume $\|z_\tau\| \leq c \|h_\tau\|$ for every $\tau \leq t$. For such τ , the decomposition $w_\tau = h_\tau + z_\tau$, the induction hypothesis and the upper bound $\|h_\tau\| \leq c_h \mu_+^\tau \ell_0$ of Lemma 11 give

$$\|w_\tau\| \leq (1+c) \|h_\tau\| \leq (1+c) c_h \mu_+^\tau \ell_0. \quad (3.32)$$

Using (3.23) to propagate each remainder, (3.15) to bound it, and then (3.32),

$$\begin{aligned} \|z_{t+1}\| &\leq \sum_{\tau=0}^t \|\mathbf{M}^{t-\tau} u_\tau\| \leq \sum_{\tau=0}^t C_{\mathbf{M}} \mu_+^{t-\tau} \chi \|w_\tau\| \\ &\leq C_{\mathbf{M}} \chi (1+c) c_h \ell_0 \sum_{\tau=0}^t \mu_+^{t-\tau} \mu_+^\tau \\ &= C_{\mathbf{M}} \chi (1+c) c_h (t+1) \mu_+^t \ell_0. \end{aligned}$$

Since $t \leq T-1$ we have $t+1 \leq T+1$, so (3.30) applies and yields

$$\|z_{t+1}\| \leq C_{\mathbf{M}} c_h (1+c) (T+1) \chi \mu_+^t \ell_0 \leq c c_{\text{esc}} \mu_+^{t+1} \ell_0 \leq c \|h_{t+1}\|,$$

the last inequality by the lower bound of Lemma 11. This closes the induction.

Conclusion. Both Δx_T and Δx_T^{hom} are the first blocks of w_T and h_T respectively, so their difference is the first block of z_T and has norm at most $\|z_T\|$. The reverse triangle inequality, (3.27) and $\|z_T\| \leq c \|h_T\| \leq c c_h \mu_+^T \ell_0$ then give

$$\|\Delta x_T\| \geq |\Delta x_T^{\text{hom}}| - \|z_T\| \geq (c_{\text{esc}} - c c_h) \mu_+^T \ell_0,$$

which is (3.31). The constraint $c < c_{\text{esc}}/c_h$ makes this bound positive. $\square$

The two trajectories therefore separate at the geometric rate μ_+ , even when neither of them escapes. Section 3.3.5 turns this into the escape guarantee.

3.3.5 Escape with high probability

We now carry out Step 3. Suppose two seeds separated by ℓ_0 along $v_{\min}$ both fail to escape. By Lemma 12 their trajectories are at least $(c_{\text{esc}} - c c_h) \mu_+^T \ell_0$ apart at the end of the window, since both stay localized; but each run remains in a ball of radius S around its own start, so they are at most $2S + \ell_0$ apart. Comparing the two bounds leaves ℓ_0 smaller than a threshold that decreases as T grows. Two seeds that both fail to escape are therefore close along $v_{\min}$, so the set of such seeds is thin in that direction, and a volume computation turns thinness into a probability.

Fix a snapshot $\tilde{x}$ with $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \leq -\varepsilon_H$, and let $\hat{x}_0, \dots, \hat{x}_T$ denote the window run from the seed ξ , that is from $\hat{x}_0 = \tilde{x} + \xi$. Define

$$\mathcal{X}_{\text{loc}} := \{\xi \in B_0(R) : \|\hat{x}_t - \hat{x}_0\| \leq S \text{ for all } 0 \leq t \leq T\}, \quad (3.33)$$

$$\mathcal{X}_{\text{stuck}} := \{\xi \in B_0(R) : f(\hat{x}_T) - f(\hat{x}_0) > -F\}. \quad (3.34)$$

By Lemma 5, a seed whose window fails to decrease f by more than F keeps every iterate within S of $\hat{x}_0$, so

$$\mathcal{X}_{\text{stuck}} \subseteq \mathcal{X}_{\text{loc}}. \quad (3.35)$$

It therefore suffices to bound the probability of landing in $\mathcal{X}_{\text{loc}}$. Finally, decompose a seed along the escape direction and its orthogonal complement,

$$\xi = \xi_1 v_{\min} + \xi_{\perp}, \quad \xi_1 = \langle \xi, v_{\min} \rangle \in \mathbb{R}, \quad \xi_{\perp} \in v_{\min}^{\perp} \cong \mathbb{R}^{n-1}. \quad (3.36)$$

Lemma 13. *Let the hypotheses of Lemma 12 hold and assume $(c_{\text{esc}} - c c_h) \mu_+^T > 1$. If ξ and $\xi' = \xi + \ell_0 v_{\min}$ both belong to $\mathcal{X}_{\text{loc}}$, then*

$$\ell_0 \leq \ell^* := \frac{2S}{(c_{\text{esc}} - c c_h) \mu_+^T - 1}. \quad (3.37)$$

Proof. The two runs start at $\hat{x}_0 = \tilde{x} + \xi$ and $\hat{x}'_0 = \tilde{x} + \xi'$, so $\|\hat{x}_0 - \hat{x}'_0\| = \ell_0$. Both seeds lie in $\mathcal{X}_{\text{loc}}$, so each run stays within S of its own start, and the triangle inequality gives

$$\|\Delta x_T\| \leq \|\hat{x}_T - \hat{x}_0\| + \|\hat{x}_0 - \hat{x}'_0\| + \|\hat{x}'_0 - \hat{x}'_T\| \leq 2S + \ell_0. \quad (3.38)$$

Both runs being localized, Lemma 12 applies and gives (3.31). Combining it with (3.38),

$$(c_{\text{esc}} - c c_h) \mu_+^T \ell_0 \leq 2S + \ell_0,$$

that is, $[(c_{\text{esc}} - c c_h) \mu_+^T - 1] \ell_0 \leq 2S$. The bracket is positive by hypothesis, so dividing by it yields (3.37). $\square$

Lemma 13 says that $\mathcal{X}_{\text{loc}}$ contains no two points more than ℓ^* apart along $v_{\min}$: it is a slab of width at most ℓ^* in that direction. The next lemma converts this into a volume bound.

Lemma 14. *Under the hypotheses of Lemma 13,*

$$\mathbb{P}(\xi \in \mathcal{X}_{\text{loc}}) \leq \frac{\ell^*}{R} \sqrt{\frac{n+1}{2\pi}}, \quad \xi \sim \text{Unif}(B_0(R)). \quad (3.39)$$

Proof. Fix $\xi_{\perp} \in v_{\min}^{\perp}$ and consider the corresponding slice of the localized set,

$$I(\xi_{\perp}) := \{a \in \mathbb{R} : a v_{\min} + \xi_{\perp} \in \mathcal{X}_{\text{loc}}\}.$$

Let $a, a' \in I(\xi_{\perp})$ with $a < a'$. The two seeds $\xi = a v_{\min} + \xi_{\perp}$ and $\xi' = a' v_{\min} + \xi_{\perp}$ satisfy $\xi' = \xi + (a' - a) v_{\min}$ and both lie in $\mathcal{X}_{\text{loc}}$, so Lemma 13 gives $a' - a \leq \ell^*$. Hence $I(\xi_{\perp})$ has diameter at most ℓ^* , and its one-dimensional Lebesgue measure satisfies $|I(\xi_{\perp})| \leq \ell^*$.

By Fubini's theorem in the coordinates (3.36), the volume of $\mathcal{X}_{\text{loc}}$ is the integral of its slice measures, so

$$\text{Vol}(\mathcal{X}_{\text{loc}}) = \int_{B_0^{(n-1)}(R)} |I(\xi_{\perp})| d\xi_{\perp} \leq \ell^* \text{Vol}(B_0^{(n-1)}(R)),$$

where $B_0^{(n-1)}(R)$ is the ball of radius R in $v_{\min}^{\perp}$.

Since ξ is uniform on $B_0(R)$,

$$\mathbb{P}(\xi \in \mathcal{X}_{\text{loc}}) = \frac{\text{Vol}(\mathcal{X}_{\text{loc}})}{\text{Vol}(B_0(R))} \leq \ell^* \frac{\text{Vol}(B_0^{(n-1)}(R))}{\text{Vol}(B_0(R))}.$$

Using $\text{Vol}(B_0^{(m)}(R)) = \pi^{m/2} R^m / \Gamma(\frac{m}{2} + 1)$,

$$\frac{\text{Vol}(B_0^{(n-1)}(R))}{\text{Vol}(B_0(R))} = \frac{1}{R\sqrt{\pi}} \cdot \frac{\Gamma(\frac{n}{2} + 1)}{\Gamma(\frac{n+1}{2})} \leq \frac{1}{R\sqrt{\pi}} \sqrt{\frac{n+1}{2}} = \frac{1}{R} \sqrt{\frac{n+1}{2\pi}},$$

where the inequality uses the property of the Gamma function that $\Gamma(x+1)/\Gamma(x+\frac{1}{2}) \leq \sqrt{x+\frac{1}{2}}$ for $x \geq 0$, applied at $x = n/2$. Combining the last two displays gives (3.39). $\square$

Lemma 15. *Let $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \leq -\varepsilon_H$, let the hypotheses of Lemma 12 hold, and let $\delta \in (0, 1)$ and $\omega \in (0, 1)$. Suppose the window length satisfies*

$$T \geq \frac{1}{\log \mu_+} \log \left[\frac{1}{c_{\text{esc}} - c c_h} \left(1 + \frac{2S}{\delta R} \sqrt{\frac{n+1}{2\pi}} \right) \right], \quad (3.40)$$

and the perturbation radius satisfies

$$\varepsilon_g R + \frac{L_1}{2} R^2 \leq \omega F, \quad \text{for which} \quad R \leq \min \left\{ \frac{\omega F}{2\varepsilon_g}, \sqrt{\frac{\omega F}{L_1}} \right\} \text{ suffices.} \quad (3.41)$$

Then, with probability at least $1 - \delta$ over $\xi \sim \text{Unif}(B_0(R))$,

$$f(\hat{x}_T) - f(\tilde{x}) \leq -(1 - \omega)F = -F_{\text{cert}}. \quad (3.42)$$

Proof. Exponentiating (3.40) gives

$$(c_{\text{esc}} - c c_h) \mu_+^T \geq 1 + \frac{2S}{\delta R} \sqrt{\frac{n+1}{2\pi}} > 1, \quad (3.43)$$

which is the hypothesis required by Lemma 13, so that both Lemma 13 and Lemma 14 apply.

We first bound the probability of landing in the stuck set. Rearranging (3.43),

$$(c_{\text{esc}} - c c_h) \mu_+^T - 1 \geq \frac{2S}{\delta R} \sqrt{\frac{n+1}{2\pi}},$$

and using Lemma 13:

$$\ell^* = \frac{2S}{(c_{\text{esc}} - c c_h) \mu_+^T - 1} \leq \delta R \sqrt{\frac{2\pi}{n+1}}.$$

Substituting this into (3.39) and using the inclusion (3.35),

$$\mathbb{P}(\xi \in \mathcal{X}_{\text{stuck}}) \leq \mathbb{P}(\xi \in \mathcal{X}_{\text{loc}}) \leq \frac{\ell^*}{R} \sqrt{\frac{n+1}{2\pi}} \leq \delta.$$

It remains to show that a seed outside the stuck set produces the announced decrease. Suppose then that $\xi \notin \mathcal{X}_{\text{stuck}}$, which by the bound just established occurs with probability at least $1 - \delta$. The escape window itself decreases f by more than F , since by (3.34)

$$f(\hat{x}_T) - f(\hat{x}_0) \leq -F,$$

while the perturbation increases it by at most ωF : applying L_1 -smoothness at $\tilde{x}$ and using $\|\xi\| \leq R$ together with $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$,

$$f(\hat{x}_0) - f(\tilde{x}) \leq \langle \nabla f(\tilde{x}), \xi \rangle + \frac{L_1}{2} \|\xi\|^2 \leq \varepsilon_g R + \frac{L_1}{2} R^2 \leq \omega F,$$

the last inequality being (3.41). Adding the two displays,

$$f(\hat{x}_T) - f(\tilde{x}) = (f(\hat{x}_T) - f(\hat{x}_0)) + (f(\hat{x}_0) - f(\tilde{x})) \leq -F + \omega F = -(1 - \omega)F,$$

which is (3.42).

Finally, the condition stated as sufficient for (3.41) is indeed so: $R \leq \omega F / (2\varepsilon_g)$ gives $\varepsilon_g R \leq \omega F / 2$, and $R \leq \sqrt{\omega F / L_1}$ gives $\frac{L_1}{2} R^2 \leq \omega F / 2$, so their sum is at most ωF . $\square$

Lemma 15 completes the escape phase: a window started at a strict saddle decreases f by at least F_{cert} with probability $1 - \delta$, which is exactly the test (3.2) performed by Algorithm 2. It remains to choose T , S , R and F so that (3.11), (3.30), (3.40) and (3.41) hold simultaneously, and to count the iterations; this is done in Section 3.4.

3.4 Iteration Complexity

In this section, we prove Theorem 1. Sections 3.2 and 3.3 each imposed conditions on the derived constants T , S , R , F and on the per-window failure probability δ . We first collect these into a single system (Section 3.4.1), then show the system is solvable and compute the resulting scales (Section 3.4.2). Finally, we count iterations against the budget Δ_f (Section 3.4.3).

The count itself is deterministic: it uses only the per-step decreases of Section 3.2 and the outcome of the certification test (3.2). Lemma 15 enters only to guarantee that a point returned by that test is second-order stationary.

3.4.1 The constraint system

We collect the conditions imposed by Section 3.3. The derived constants must satisfy the following, where the reference after each line gives its origin.

$$F = L_1 \left(\frac{1}{2} - \gamma \right) \frac{S^2}{T}, \quad (3.11), \text{ solved for } F \quad (3.44)$$

$$\chi = \frac{\sqrt{2} L_2 (S + R)}{L_1} \leq \frac{\Lambda}{T + 1}, \quad \Lambda := \frac{c c_{\text{esc}} \mu_+}{C_{\mathbf{M}} c_h (1 + c)}, \quad (3.30), (3.15) \quad (3.45)$$

$$\varepsilon_g R + \frac{L_1}{2} R^2 \leq \omega F, \quad (3.41) \quad (3.46)$$

$$T \geq \frac{1}{\log \mu_+} \log \left[\frac{1}{c_{\text{esc}} - c c_h} \left(1 + \frac{2S}{\delta R} \sqrt{\frac{n+1}{2\pi}} \right) \right]. \quad (3.40) \quad (3.47)$$

Here, $c \in (0, \min\{1, c_{\text{esc}}/c_h\})$ and $\omega \in (0, 1)$ are free parameters of the analysis, and c_{esc} , c_h , $C_{\mathbf{M}}$, μ_+ are the spectral constants of Lemmas 10 and 11, depending only on γ and ε_H .

3.4.2 Solving the system

We proceed in two stages. First we express S , F and R as functions of the window length T , in such a way that (3.44)–(3.46) hold for *every* $T \geq 1$ (Lemma 16). This leaves only (3.47), which becomes a single inequality in T ; we then check that it holds for all T large enough (Proposition 1). Throughout we write

$$\kappa_\gamma := L_1 \left(\frac{1}{2} - \gamma \right) > 0.$$

Lemma 16. *For $T \geq 1$ define*

$$\Psi(T) := \frac{\Lambda L_1}{\sqrt{2} L_2 (T+1)}, \quad S(T) := \frac{\Psi(T)}{2}, \quad F(T) := \kappa_\gamma \frac{S(T)^2}{T}, \quad (3.48)$$

$$R(T) := \min \left\{ \frac{\omega F(T)}{2\varepsilon_g}, \sqrt{\frac{\omega F(T)}{L_1}} \right\}. \quad (3.49)$$

Then $S(T), F(T), R(T) > 0$, and (3.44), (3.45) and (3.46) hold for every $T \geq 1$.

Proof. Positivity is trivial, and (3.44) holds by definition.

(3.46): the two entries of the minimum in (3.49) control one term each. From $R \leq \omega F / (2\varepsilon_g)$ we get $\varepsilon_g R \leq \frac{1}{2} \omega F$, and from $R \leq \sqrt{\omega F / L_1}$ we get $\frac{L_1}{2} R^2 \leq \frac{1}{2} \omega F$. Adding gives $\varepsilon_g R + \frac{L_1}{2} R^2 \leq \omega F$.

(3.45): it says $S + R \leq \Psi$, and since $S = \Psi/2$ this amounts to $R \leq S$. Substituting $F = \kappa_\gamma S^2 / T$ and $\kappa_\gamma / L_1 = \frac{1}{2} - \gamma$ into the second entry of (3.49),

$$\sqrt{\frac{\omega F(T)}{L_1}} = S(T) \sqrt{\frac{\omega (\frac{1}{2} - \gamma)}{T}} \leq S(T), \quad (3.50)$$

because $\omega < 1$, $\frac{1}{2} - \gamma < \frac{1}{2}$ and $T \geq 1$. As R is a minimum, $R(T) \leq S(T)$, so $S + R \leq 2S = \Psi$. $\square$

It remains to satisfy (3.47). Substituting (3.48) and (3.49) turns its right-hand side into a function of T alone,

$$G(T) := \frac{1}{\log \mu_+} \log \left[\frac{1}{c_{\text{esc}} - c_{ch}} \left(1 + \frac{2S(T)}{\delta R(T)} \sqrt{\frac{n+1}{2\pi}} \right) \right], \quad (3.51)$$

so that (3.47) reads $T \geq G(T)$.

Proposition 1. *Let $\gamma \in (0, \frac{1}{2})$, $c \in (0, \min\{1, c_{\text{esc}}/c_h\})$, $\omega \in (0, 1)$, $\varepsilon_g \in (0, 1]$, $\varepsilon_H \in (0, L_1]$ and $\delta \in (0, 1)$. Then $T \geq G(T)$ holds for all T large enough. For any such T , the constants $S(T), F(T), R(T)$ of (3.48)–(3.49) satisfy the full system (3.44)–(3.47).*

Proof. By Lemma 16 only (3.47), i.e. $T \geq G(T)$, remains, so it suffices to show that G grows logarithmically.

The variable T enters G only through the ratio S/R . Inverting the minimum in (3.49) and substituting $F = \kappa_\gamma S^2 / T$,

$$\frac{S(T)}{R(T)} = \max \left\{ \frac{2\varepsilon_g T}{\omega \kappa_\gamma S(T)}, \sqrt{\frac{T}{\omega (\frac{1}{2} - \gamma)}} \right\}. \quad (3.52)$$

By (3.48), $1/S(T)$ is a linear function of T , so the first entry is a quadratic polynomial in T and the second is of order $\sqrt{T}$. Hence $S(T)/R(T) \leq P(T)$ for some quadratic polynomial P with positive coefficients depending only on $\gamma, c, \omega, L_1, L_2$ and ε_g . The bracket in (3.51) is therefore at most a quadratic polynomial in T , and

$$G(T) = \mathcal{O}(\log T) \quad \text{as } T \rightarrow \infty.$$

Since $T - G(T) \rightarrow +\infty$, there is $T_0 \geq 1$ with $T \geq G(T)$ for all $T \geq T_0$, which is (3.47). $\square$

Lemma 17. Let $\gamma \in (0, \frac{1}{2})$, $\alpha_w = 1/L_1$, $\varepsilon_H \in (0, L_1]$, and let $\lambda_{\min} \leq -\varepsilon_H$. Set $a := \alpha_w |\lambda_{\min}|$. Then $a \in [\alpha_w \varepsilon_H, 1]$ and

$$1 + a \leq \mu_+ < 1 + \frac{a}{1-\gamma} < 3, \quad \frac{a}{2} \leq \log \mu_+ < \frac{a}{1-\gamma}. \quad (3.53)$$

In particular $\log \mu_+ \geq \varepsilon_H/(2L_1)$.

Proof. By Assumption 1, $|\lambda_{\min}| \leq \|H\| \leq L_1$, so $a \leq 1$; and $a \geq \alpha_w \varepsilon_H$ by hypothesis.

Factoring $p(\mu) = (\mu - \mu_+)(\mu - \mu_-)$ and evaluating at 1 gives $(1 - \mu_+)(1 - \mu_-) = p(1) = 1 - t + \gamma = -a$, that is

$$(\mu_+ - 1)(1 - \mu_-) = a.$$

By Lemma 8, $\mu_+ > 1$, so $\mu_- = \gamma/\mu_+ \in (0, \gamma)$ and hence $1 - \gamma < 1 - \mu_- < 1$. Dividing,

$$a < \mu_+ - 1 < \frac{a}{1-\gamma},$$

which with $a \leq 1$ and $\frac{1}{1-\gamma} < 2$ gives the first chain in (3.53).

For the logarithm, $\log \mu_+ \leq \mu_+ - 1 < a/(1-\gamma)$. Conversely $\log \mu_+ \geq \log(1+a) \geq a/2$, since $h(u) = \log(1+u) - u/2$ satisfies $h(0) = 0$ and $h'(u) = (1-u)/(2(1+u)) \geq 0$ on $[0, 1]$. Finally $a \geq \alpha_w \varepsilon_H = \varepsilon_H/L_1$. $\square$

The rate μ_+ depends on the curvature $|\lambda_{\min}|$ at the snapshot, which is not known when the parameters are set. All constants are therefore chosen for the worst admissible case $|\lambda_{\min}| = \varepsilon_H$, using the guaranteed bound $\log \mu_+ \geq \varepsilon_H/(2L_1)$. A snapshot with more negative curvature only escapes faster.

Proposition 1 settles existence: Algorithm 2 can be run with parameters meeting every requirement of Section 3.3. Computing a complexity bound requires their size as well. We therefore take T to be the smallest admissible window length and evaluate the resulting orders of magnitude.

Proposition 2. Let T be the smallest integer satisfying (3.47), let S, F, R be as in (3.48)–(3.49), and set

$$\phi := 1 + \log \left(\frac{n L_1 L_2 \Delta_f}{\varepsilon_H \delta} \right).$$

Then

$$T = \mathcal{O} \left(\frac{L_1 \phi}{\varepsilon_H} \right), \quad S = \Omega \left(\frac{\varepsilon_H}{L_2 \phi} \right), \quad F = \Omega \left(\frac{\varepsilon_H^3}{L_2^2 \phi^3} \right),$$

the implied constants depending only on γ, c and ω . All three hold with Θ in place of $\mathcal{O}$ and Ω when $|\lambda_{\min}| = \Theta(\varepsilon_H)$.

Proof. We begin by checking that the constants appearing in (3.45) are of order one. By Lemma 17, $\mu_+ \in (1, 3)$, so $c_{\text{esc}} = (\mu_+ - \gamma)/(\mu_+ - \mu_-) > (1 - \gamma)/3$; and by Lemmas 10 and 11, c_h and C_M are bounded above by expressions involving γ alone. Hence $\Lambda = \Theta(1)$, with implied constants depending only on γ and c .

We turn to the window length, which is governed by the rate $\log \mu_+$ appearing in (3.51). Lemma 17 bounds it from below,

$$\log \mu_+ \geq \frac{\varepsilon_H}{2L_1}, \quad (3.54)$$

and it remains to control the logarithm in (3.51). By (3.52), the ratio $S(T)/R(T)$ is at most a quadratic polynomial in T whose coefficients are polynomial in L_1, L_2, ε_g and the constants above; since $\varepsilon_g \leq 1$, the bracket in (3.51) is then polynomial in $n, L_1, L_2, 1/\delta$ and T , so its logarithm is $\mathcal{O}(\phi + \log T)$. Combining this with (3.54),

$$G(T) = \mathcal{O} \left(\frac{L_1}{\varepsilon_H} (\phi + \log T) \right).$$

The smallest T satisfying $T \geq G(T)$ is therefore polynomially bounded in the problem data, hence $\log T = \mathcal{O}(\phi)$ and

$$T = \mathcal{O}\left(\frac{L_1 \phi}{\varepsilon_H}\right).$$

The remaining constants follow from this bound. By (3.48), $S = \Psi/2 = \Theta(L_1/(L_2 T))$, so an upper bound on T becomes a lower bound on S ,

$$S = \Omega\left(\frac{\varepsilon_H}{L_2 \phi}\right),$$

and substituting into $F = \kappa_\gamma S^2/T$, using the bound on T once more,

$$F = \Omega\left(L_1 \cdot \frac{\varepsilon_H^2}{L_2^2 \phi^2} \cdot \frac{\varepsilon_H}{L_1 \phi}\right) = \Omega\left(\frac{\varepsilon_H^3}{L_2^2 \phi^3}\right).$$

Finally, these bounds are two-sided when the curvature at the snapshot is of order ε_H . Indeed, if $|\lambda_{\min}| = \Theta(\varepsilon_H)$ then $a = \Theta(\varepsilon_H/L_1)$, so $\log \mu_+ = \Theta(\varepsilon_H/L_1)$ by (3.53); the bound on T is then attained, and the two inequalities above become equalities in order. $\square$

Remark 6. The window length $T = \mathcal{O}(L_1 \phi / \varepsilon_H)$ of Proposition 2 is not an artefact of the analysis. By Lemma 17, $\mu_+ < 1 + a/(1 - \gamma)$ with $a = \alpha_w |\lambda_{\min}|$, so for small a the escape rate improves on the gradient-descent rate $1 + a$ only by the constant factor $1/(1 - \gamma)$: the dependence on a is unchanged. At $\gamma = 1$ the characteristic polynomial $\mu^2 - (1 + \gamma + a)\mu + \gamma$ instead gives $\mu_+ \approx 1 + \sqrt{a}$, the accelerated rate, but the energy (3.8) ceases to be monotone at $\gamma \geq \frac{1}{2}$, so the localization of Lemma 5 is unavailable exactly where acceleration would begin. A constant momentum bounded away from 1 therefore cannot shorten the window.

3.4.3 Counting the iterations

Each escape window that does not return an $(\varepsilon_g, \varepsilon_H)$ -SOSP decreases f by at least F_{cert} , so the number of windows the algorithm can run is bounded by the budget Δ_f . Write $\bar{K}_{\max}$ for that bound and δ for the failure probability allotted to a single window, so that the total failure probability is δ_{tot} :

$$\bar{K}_{\max} := \frac{\Delta_f}{(1 - \omega)F} + 1, \quad \delta := \frac{\delta_{\text{tot}}}{\bar{K}_{\max}}. \quad (3.55)$$

These two definitions are circular as they stand: $\bar{K}_{\max}$ depends on F , which by Proposition 2 depends on ϕ , which depends on $\log(1/\delta)$. We break the circularity by fixing

$$\phi_0 := 1 + \log\left(\frac{n L_1 L_2 \Delta_f}{\varepsilon_H \delta_{\text{tot}}}\right)$$

in advance and evaluating T, S, F, R with ϕ_0 in place of ϕ . Since $\bar{K}_{\max}$ is then polynomial in the problem data, $\log(1/\delta) = \mathcal{O}(\phi_0)$, so the bounds of Proposition 2 hold with ϕ_0 .

Proof of Theorem 1. Fix the parameters as in Proposition 1, with δ as in (3.55).

We first record the decrease available at each type of iteration. From Algorithm 2, a descent step has $\|g_k\| > \varepsilon_g$. If $k \in \mathcal{N}$, then (3.3) together with $\varepsilon_g \leq 1$ and $1 + p - q \geq 0$ gives

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{N}} \varepsilon_g^{\max\{1+p, 2(1+p-q)\}},$$

while if $k \in \mathcal{R}$, then (3.4) gives $f(x_k) - f(x_{k+1}) > c_{\mathcal{R}} \varepsilon_g^2$. In both cases the decrease is strictly positive.

Escape windows contribute in the same way, provided they do not terminate the algorithm. Index the perturbation episodes by $j = 1, 2, \dots$; episode j takes a snapshot $\tilde{x}^{(j)}$ and runs T

window steps, ending at $\hat{x}_T^{(j)}$. If the certification (3.2) does not return an $(\varepsilon_g, \varepsilon_H)$ -SOSP at the end of episode j , then by the test itself

$$f(\tilde{x}^{(j)}) - f(\hat{x}_T^{(j)}) \geq F_{\text{cert}} = (1 - \omega)F > 0.$$

These decreases can now be set against the total budget. They are nonnegative, and they are attached to disjoint phases of the run: a descent step k accounts for the passage from x_k to x_{k+1} , and an episode j for the passage from $\tilde{x}^{(j)}$ to $\hat{x}_T^{(j)}$, so that no iteration belongs to two such phases. Their sum therefore telescopes to at most $f(x_0) - f_{\text{low}} = \Delta_f$, therefore

$$|\mathcal{N}| \leq \frac{\Delta_f}{c_{\mathcal{N}}} \varepsilon_g^{-\max\{1+p, 2(1+p-q)\}}, \quad |\mathcal{R}| \leq \frac{\Delta_f}{c_{\mathcal{R}}} \varepsilon_g^{-2}, \quad \bar{K} \leq \bar{K}_{\max}, \quad (3.56)$$

where $\bar{K}$ is the number of episodes actually executed and $\bar{K}_{\max}$ is as in (3.55).

Every iteration of Algorithm 2 lies in either $\mathcal{N}$, $\mathcal{R}$ or $\mathcal{W}$, and $|\mathcal{W}| \leq T\bar{K} \leq T\bar{K}_{\max}$ by (3.56), so all three counts are finite and the algorithm performs finitely many iterations. Its outer loop being infinite, and exiting only through the **return** of the certification test, that test must fire; the algorithm therefore returns some $\tilde{x}$.

It remains to convert the episode count into a number of window iterations. By Proposition 2, $F = \Omega(\varepsilon_H^3 / (L_2^2 \phi_0^3))$ and $T = \mathcal{O}(L_1 \phi_0 / \varepsilon_H)$, so

$$\bar{K}_{\max} = \mathcal{O}\left(\frac{L_2^2 \Delta_f \phi_0^3}{\varepsilon_H^3}\right), \quad |\mathcal{W}| \leq T \bar{K}_{\max} = \mathcal{O}\left(\frac{L_1 L_2^2 \Delta_f \phi_0^4}{\varepsilon_H^4}\right) = \tilde{\mathcal{O}}(\varepsilon_H^{-4}),$$

since ϕ_0 is logarithmic in n , $1/\delta_{\text{tot}}$ and the problem constants.

Everything so far has been deterministic. Randomness enters only in establishing that the point returned by the certification test really is second-order stationary. For each episode j let

$$\mathcal{E}_j := \{\lambda_{\min}(\nabla^2 f(\tilde{x}^{(j)})) \leq -\varepsilon_H \text{ and the certification returns } \tilde{x}^{(j)}\}$$

be the event that episode j returns a point which is in fact a strict saddle. On $\{\lambda_{\min}(\nabla^2 f(\tilde{x}^{(j)})) \leq -\varepsilon_H\}$ the snapshot satisfies the hypotheses of Lemma 15, and returning an $(\varepsilon_g, \varepsilon_H)$ -SOSP means $f(\tilde{x}^{(j)}) - f(\hat{x}_T^{(j)}) < F_{\text{cert}}$, which Lemma 15 rules out except on an event of probability at most δ ; hence $\mathbb{P}(\mathcal{E}_j) \leq \delta$. Since at most $\bar{K}_{\max}$ episodes occur, the union bound and (3.55) give

$$\mathbb{P}\left(\bigcup_{j \leq \bar{K}_{\max}} \mathcal{E}_j\right) \leq \bar{K}_{\max} \delta = \delta_{\text{tot}}.$$

We conclude on the complement of $\bigcup_j \mathcal{E}_j$, an event of probability at least $1 - \delta_{\text{tot}}$. There, the returned point $\tilde{x}$ satisfies $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$, since it is a snapshot, and $\lambda_{\min}(\nabla^2 f(\tilde{x})) > -\varepsilon_H$, since otherwise some $\mathcal{E}_j$ would have occurred; it is therefore an $(\varepsilon_g, \varepsilon_H)$ -SOSP. Summing the three counts of (3.56) with the bound on $|\mathcal{W}|$ obtained above,

$$N = |\mathcal{N}| + |\mathcal{R}| + |\mathcal{W}| \leq \frac{\Delta_f}{c_{\mathcal{R}}} \varepsilon_g^{-2} + \frac{\Delta_f}{c_{\mathcal{N}}} \varepsilon_g^{-\max\{1+p, 2(1+p-q)\}} + \mathcal{O}\left(\frac{L_1 L_2^2 \Delta_f \phi_0^4}{\varepsilon_H^4}\right),$$

which is the bound of Theorem 1. □

Chapter 4

Numerical Experiments

This chapter reports experiments conducted with an implementation of Algorithm 2. Section 4.2 computes the constants of (3.44)–(3.47) $(T, F, R, \bar{K})$, shows that the values prescribed by the analysis are too conservative to run, and proposes a practical alternative. Section 4.3 isolates the descent phase and compares safeguarded NCG against GD. Section 4.4 compares the full method with PGD [29] and PAGD [30], in iterations and in oracle evaluations. Section 4.5 is an ablation study, it varies the escape window length (T) and the algorithm used by the escape (HB vs GD vs AGD). Additionally, we study the use of a negative-curvature oracle instead of a first-order escape window. Section 4.6 lists the limitations of the analyses.

4.1 Experimental Design

We use two strict-saddle instances, both with closed-form minimizers and analytically certified constants. The first is a quadratic saddle:

$$f(x) = \frac{1}{2}x^\top Hx + \frac{\nu}{4}\|x\|^4, \quad H = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_1 = -\lambda_- < 0, \quad (4.1)$$

with minimizers $x^* = \pm\sqrt{\lambda_-/\nu}e_1$ and $f^* = f_{\text{low}} = -\lambda_-^2/(4\nu)$. Its spectrum is an input, so ε_H , μ_+ and the point $\lambda^* = L_1(1 - \sqrt{\gamma})^2$ at which $\mu_+ = \mu_-$ are known exactly. We take $n = 20$, $\lambda_- = 1/2$, $\nu = 1/10$ and $\lambda_2, \dots, \lambda_n \sim \text{Unif}[\frac{1}{2}, 1]$, redrawn per repetition.

The second is symmetric low-rank matrix factorization:

$$f(U) = \frac{1}{4}\|UU^\top - M\|_F^2, \quad M = U^*(U^*)^\top \in \mathbb{R}^{d \times d} \text{ of rank } r, \quad (4.2)$$

with $n = dr$, every local minimizer global ($f^* = f_{\text{low}} = 0$) and all saddles strict [16, 17]. Write $\sigma_1 \geq \dots \geq \sigma_r > 0$ for the nonzero eigenvalues of M ; the origin is then a strict saddle with $\lambda_{\min} = -\sigma_1$. Planting $U^* = Q \text{diag}(\sqrt{\sigma_1}, \dots, \sqrt{\sigma_r})$ with $Q \in \mathbb{R}^{d \times r}$ having orthonormal columns gives M the spectrum we want. We take $d = 10$, $r = 7$ and $\sigma_1, \dots, \sigma_r$ evenly spaced on $[1, 3]$ for numerics.

Both objectives are of degree four, so global Lipschitz constants do not exist and Assumptions 1 and 2 are certified on a working ball $\|x\| \leq B_{\text{dom}}$, taken as twice the norm of the known minimizers. This gives $L_1 = \max_i |\lambda_i| + 3\nu B_{\text{dom}}^2$ and $L_2 = 6\nu B_{\text{dom}}$ for (4.1), and $L_1 = 3B_{\text{dom}}^2 + \|M\|_2$, $L_2 = 6B_{\text{dom}}$ for (4.2). The resulting Lipschitz constants are conservative, which inflates the window length T ; Table 4.1 quantifies this.

Starting points The two groups of experiments use different starts.

For the descent-phase experiments of Section 4.3, starts are generic and the perturbation mechanism is switched off, so that the measured quantity is the descent iteration alone.

The escape experiments of Sections 4.4 and 4.5 start at a saddle. From a generic initialization a strict saddle is avoided almost surely and no perturbation is ever triggered, so nothing would

be measured. For (4.1) we take x_0 to be a small Gaussian with its components along the negative eigenvectors removed. For (4.2) we take the columns of U_0 in the orthogonal complement of $\text{range}(M)$, which requires $d > r$: then $MU_0 = 0$, so the gradient is negligible while $\lambda_{\min} \approx -\sigma_1(M)$. What follows therefore measures how the escape behaves once a saddle is reached, not how often one is reached.

Methods and parameters All methods share the stepsize $\alpha_w = 1/L_1$ and, where applicable, the escape budget (T, R, F_{cert}) ; no parameter is tuned per method or per instance. Besides GD we run PGD [29] and PAGD [30], both given the same escape budget. For PAGD we keep $\eta_{\text{PAGD}} = 1/L_1$ and rescale $\theta_{\text{PAGD}} = \min\{\frac{1}{2}, 1/(2\sqrt{L_1/\varepsilon_H})\}$ so that its exploitation threshold matches the reference configuration. We run Algorithm 2 with the PRP+ beta, and Algorithm 2 with the perturbations disabled, referred to below as *safeguarded NCG*.

We also run *perturbed NCG*, the NCG analogue of PGD and PAGD: the same perturb-and-certify loop and the same budget (T, R, F_{cert}) , but the window continues with NCG steps at the constant stepsize $\alpha_w = 1/L_1$ and with the restart test suspended. It differs from Algorithm 2 in the momentum coefficient alone, and so isolates the choice of heavy ball.

Lastly, we report an optional variant of Algorithm 2, the *early exit*. A window normally runs all T steps; here it stops at the first step satisfying

$$f(x_k) - f(\tilde{x}) \leq -F_{\text{cert}} \quad \text{and} \quad \|g_k\| > \varepsilon_g, \quad (4.3)$$

handing control back to the descent phase, so that a successful escape costs fewer than T iterations. The test (3.2) is deterministic, and a *returned* SOSP still requires a full *failing* window, so the $(\varepsilon_g, \varepsilon_H)$ -SOSP certificate is untouched. Both conditions are needed: F_{cert} is a certification threshold, not a per-step progress signal, so exiting on the decrease alone would return to the descent phase while the gradient is still below ε_g , triggering a fresh perturbation at once and giving a cycle of length-one windows.

The restart test (2.6) is run with

$$\sigma = 10^{-2}, \quad \kappa = 10^2, \quad p = \frac{1}{2}, \quad q = 1, \quad (4.4)$$

and the Armijo search with $\eta = 10^{-4}$, backtracking factor $\theta = 1/2$ and at most 200 backtracks. The exponents place the descent bound $\varepsilon_g^{-\max\{1+p, 2(1+p-q)\}}$ of Theorem 1 at $\varepsilon_g^{-3/2}$, inside the regime where it improves on the ε_g^{-2} of a restart step; the natural-looking $p = q = 1$ would give exactly 2 and no improvement. The scales σ and κ are chosen so that the safeguard never triggers. With $q > p$, Lemma 3 places a gradient floor at $(\sigma/\kappa)^{1/(q-p)} = 10^{-8}$, and $d = -g$ passes the first condition of (2.6) whenever $\|g\| \geq \sigma^{1/(1-p)} = 10^{-4}$, both below $\varepsilon_g = 10^{-3}$. Across every run reported here the test never fires, so all recorded descent steps are accepted conjugate-gradient steps.

The remaining tolerances are $\varepsilon_g = 10^{-3}$, $\delta_{\text{tot}} = 0.1$, $\Delta_f = f(x_0) - f_{\text{low}}$ and $\gamma = 0.45$, just below the $\gamma < \frac{1}{2}$ required by Lemma 4. The curvature tolerance is oracle-derived,

$$\varepsilon_H = 0.8 |\lambda_{\min}(\nabla^2 f(\text{saddle}))|, \quad (4.5)$$

the factor 0.8 ensuring that the planted saddle violates (1.2) at the declared tolerance, so that returning it would be a genuine false certification.

Protocol Every reported number is a median over 20 independent repetitions, each with different seeds, starting points and perturbation draws. Runs are capped at $3 \cdot 10^4$ iterations.

We distinguish throughout between *reaching* an SOSP and *returning* one. Reaching is the index of the first $(\varepsilon_g, \varepsilon_H)$ -SOSP along the trajectory, certified offline with the true Hessian, while returning is the iteration at which the algorithm itself stops and certifies. The two differ because certification requires a full failing window, and the gap between them is studied in Section 4.4.

4.2 Choosing the constants

Solving (3.44)–(3.47) produces constants that cannot be used in practice. At the reference configuration for the quadratic saddle problem ($n = 20$, $L_1 = 7$, $L_2 = 2.7$, $\varepsilon_g = 10^{-3}$, $\varepsilon_H = 0.4$, $\gamma = 0.45$), it gives

$$T = 636, \quad R \approx 2 \cdot 10^{-14}, \quad F \approx 9 \cdot 10^{-17}, \quad \bar{K} \approx 2 \cdot 10^{16} :$$

The perturbation radius is smaller than the arithmetic precision of the machine, and the certification threshold is even smaller than that, so running the algorithm with these values would measure numerical error rather than the behaviour of the method. The runs therefore use the Θ -scales of Proposition 2 with the polylogarithmic factors set to one,

$$\alpha_w = \frac{1}{L_1}, \quad S = \frac{\varepsilon_H}{L_2}, \quad F = \frac{L_1(\frac{1}{2} - \gamma)S^2}{T}, \quad R = \min \left\{ \frac{\omega F}{2\varepsilon_g}, \sqrt{\frac{\omega F}{L_1}}, S \right\}, \quad F_{\text{cert}} = (1 - \omega)F, \quad (4.6)$$

with $\omega = \frac{1}{2}$ and T the smallest integer satisfying (3.47), so that the escape probability of Lemma 15 still holds. The one constraint we drop is (3.45), whose constant Λ carries the accumulated worst-case constants of the induction of Lemma 12. The runs below are therefore *not* certified by Theorem 1; they test whether the mechanism behaves as the analysis predicts at scales that can actually be run.

instance	n	L_1	L_2	$ \lambda_{\min} $	ε_H	L_1/ε_H	T	F_{cert}
quadratic saddle	20	6.97	2.68	0.50	0.40	17	208	$1.9 \cdot 10^{-5}$
matrix factorization	70	171	44.9	3.00	2.40	71	952	$1.3 \cdot 10^{-5}$

Table 4.1: Constants used for each instance. The window length T grows with L_1/ε_H , and L_1 is large mainly because it is a bound valid over the whole working ball.

4.3 The descent phase

The safeguarded descent is far more efficient than a gradient step. The iterations to $\|g\| \leq \varepsilon_g$ of the descent phase is reported in Table 4.2.

ε_g	quadratic saddle				matrix factorization			
	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-2}	10^{-4}	10^{-6}	10^{-8}
GD	109	144	174	204	1594	1989	2469	3234
safeguarded NCG, FR	4	17	21	27	83	102	108	117
safeguarded NCG, PRP+	4	15	20	26	20	26	35	47

Table 4.2: Iterations to $\|g\| \leq \varepsilon_g$ of GD, safeguarded NCG using the FR beta, and safeguarded NCG using the PRP+ beta. Safeguarded NCG with PRP+ converges the fastest to the ε_g -FOSP.

The restart test (2.6) never fires: with the exponents of (4.4) every recorded step is an accepted conjugate step, so what is measured is the conjugate direction itself and not a disguised GD. This is also the only place where the two β formulas are compared; PRP+ is the better of the two (47 against 117 on matrix factorization), and is used throughout the rest of the chapter.

The bound of Theorem 1 predicts a difference in rate, $\varepsilon_g^{-3/2}$ against ε_g^{-2} , so the ratio between the two counts ought to grow without bound as ε_g falls. It does not: across the four tolerances that ratio is 27, 10, 9, 8 on the quadratic saddle and 80, 77, 71, 69 on matrix factorization, which is flat or shrinking. Neither method comes close to its worst-case rate here: GD needs barely twice as many iterations at $\varepsilon_g = 10^{-8}$ as at 10^{-2} , whereas an ε_g^{-2} rate over that range would

demand a factor 10^{12} . Both bounds are therefore far from tight on these instances. What the table reports is therefore a large constant-factor advantage of the conjugate direction, not the asymptotic separation that the exponents (4.4) were chosen to permit; exhibiting the latter would require an instance on which GD actually attains its ε_g^{-2} worst case.

4.4 The escape phase

We now compare the full method against GD, PGD, PAGD and perturbed NCG, from a start on the saddle. Table 4.3 reports the cost of reaching the first SOSP and of returning one, in iterations and in oracle evaluations.

method	quadratic saddle			matrix factorization		
	reach	evals	return	reach	evals	return
GD	—	—	n/a	6248	6249	n/a
perturbed NCG	175	181	422	970	1015	1921
PGD	210	213	450	2181	2187	3119
PAGD	149	596	444	570	2280	1904
Algorithm 2	99	105	422	956	967	1909
Algorithm 2, early exit	50	68	256	137	203	1087

Table 4.3: Iterations and oracle evaluations (gradient plus function) to *reach* the first $(\varepsilon_g, \varepsilon_H)$ -SOSP, and iterations to *return* one. Medians over 20 repetitions. “—”: no SOSP reached within the cap on any repetition. “n/a”: no certification mechanism, GD never certifies.

GD alone never reaches an SOSP on the quadratic saddle, and needs 6248 iterations on matrix factorization. Among the perturbed methods, the comparison depends on whether we count reaching or returning.

In *reaching*, Algorithm 2 wins on the quadratic saddle (99 against PAGD’s 149) but loses on matrix factorization (956 against 570). The loss comes from the window length, not from the descent phase. On matrix factorization the window opens at the first iteration, and the decrease certifying the escape arrives after a median of 113 of its 952 steps (Figure 4.1, right panel), but the first iterate that is genuinely an SOSP appears only at 956, just after the window closes. Everything in between is spent inside a window that has already done its job. The corresponding numbers on the quadratic saddle are 33.5 of 208 (Figure 4.1, left panel), against a reach of 99. The early exit removes precisely those steps: the reach falls from 956 to 137 on matrix factorization, and from 99 to 50 on the quadratic saddle.

Perturbed NCG answers a narrower question: at the fixed stepsize, is a constant momentum coefficient the right choice inside the window? The answer is that it is at least as good for our instances. Algorithm 2 is clearly better on the quadratic saddle (99 against 175 in reaching, 105 against 181 in evaluations) and equivalent on matrix factorization (956 against 970). The choice of beta is crucial when using perturbed NCG. With the Fletcher–Reeves beta (not shown in the table) the window diverges. Since β^{FR} is unbounded and the restart test is suspended inside the window, nothing prevents the conjugate direction from growing without limit at a fixed stepsize.

In *returning*, the differences almost disappear, because every perturbed method must first spend one full window that *fails*, and that cost is the same for all of them. PAGD returns at 1904 and Algorithm 2 at 1909 on matrix factorization, whereas in reach PAGD is almost twice as fast as Algorithm 2. The early exit shortens only windows that succeed, and returning requires a window that fails, so it helps far less here than in reaching.

The evaluation counts tell a second story, visible in Figure 4.2. One PAGD iteration costs four evaluations, two gradient and two function, so wherever PAGD leads in iterations it trails in evaluations: 2280 against 967 for Algorithm 2 on matrix factorization, even before the early exit. Algorithm 2 spends one gradient per window step and nothing on the restart test, and

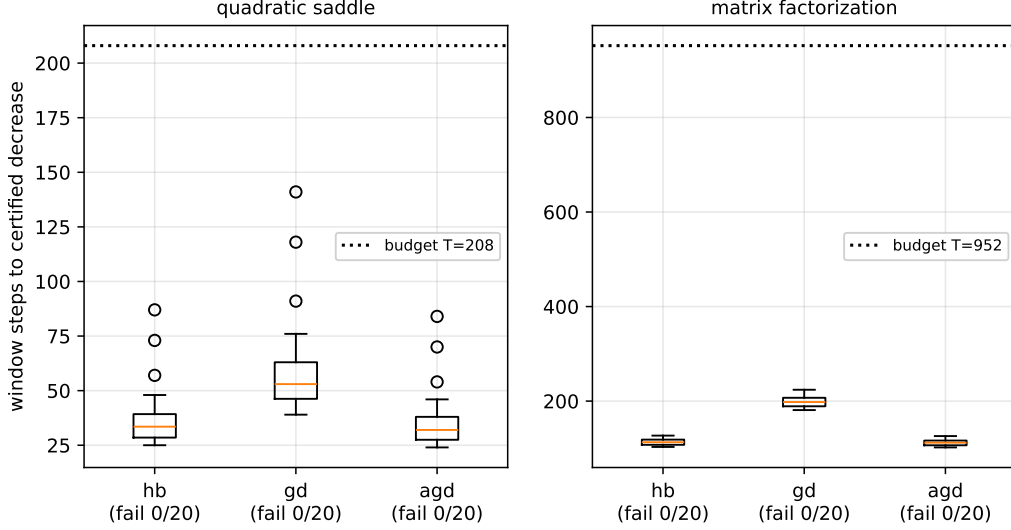


Figure 4.1: Escape time of a single window, by subroutine, over 20 repetitions per instance, at $\gamma = 0.45$ and perturbation radius $R = 1.6 \cdot 10^{-3}$ (quadratic saddle) and $1.8 \cdot 10^{-4}$ (matrix factorization). Dotted line: the window budget T . Medians are 33.5, 32.0 and 53.0 on the quadratic saddle and 113, 112 and 198 on matrix factorization, for HB, AGD and GD respectively.

although the Armijo line search costs extra function values, the runs here are dominated by window steps, so few line searches are performed and the total stays near one evaluation per iteration (1.01 and 1.07). The early exit hands control back to the descent sooner and therefore buys more line searches, which is where that cost becomes visible (1.37 and 1.48), still far below PAGD.

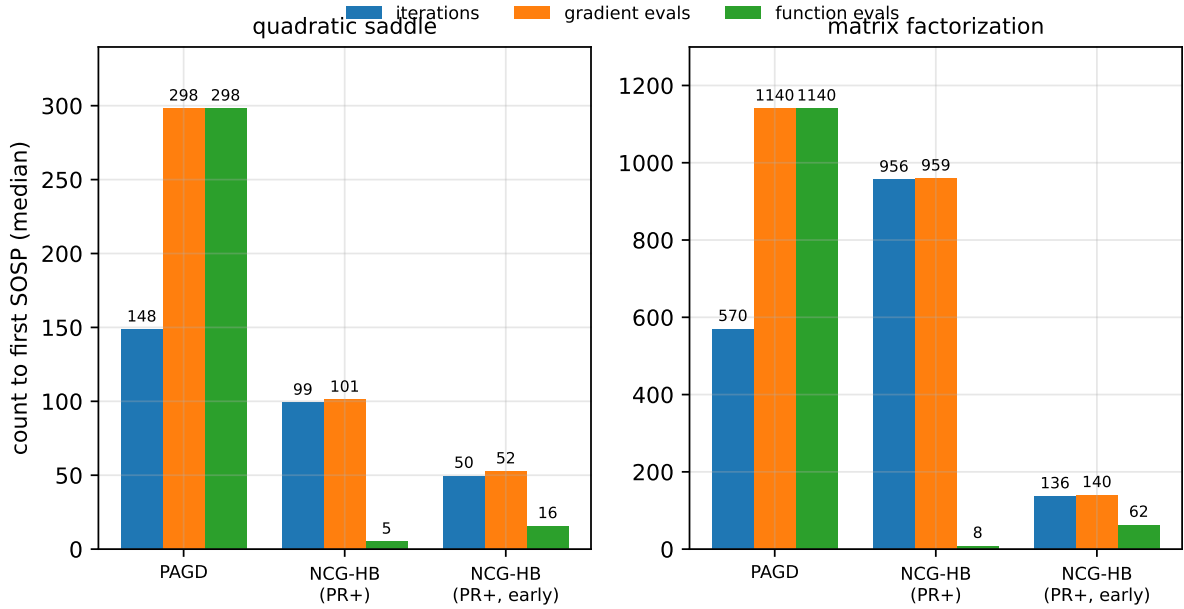


Figure 4.2: Iterations, gradient evaluations and function evaluations to the first SOS, medians over 20 repetitions. The figure separates the two: PAGD's function evaluations match its gradient evaluations one for one, while those of Algorithm 2, spent only in the line search of the descent phase, remain much smaller.

Figure 4.3 states the same comparison without reference to a median: for every budget K , the fraction of repetitions that have delivered a certified SOSP within K . A curve lying to the left is better *at every budget*, so the ranking does not depend on where one chooses to stop. The early-exit variant is leftmost in all four panels, and the second row, the same profile counted in oracle evaluations, moves PAGD from second-best to worst on the quadratic saddle.

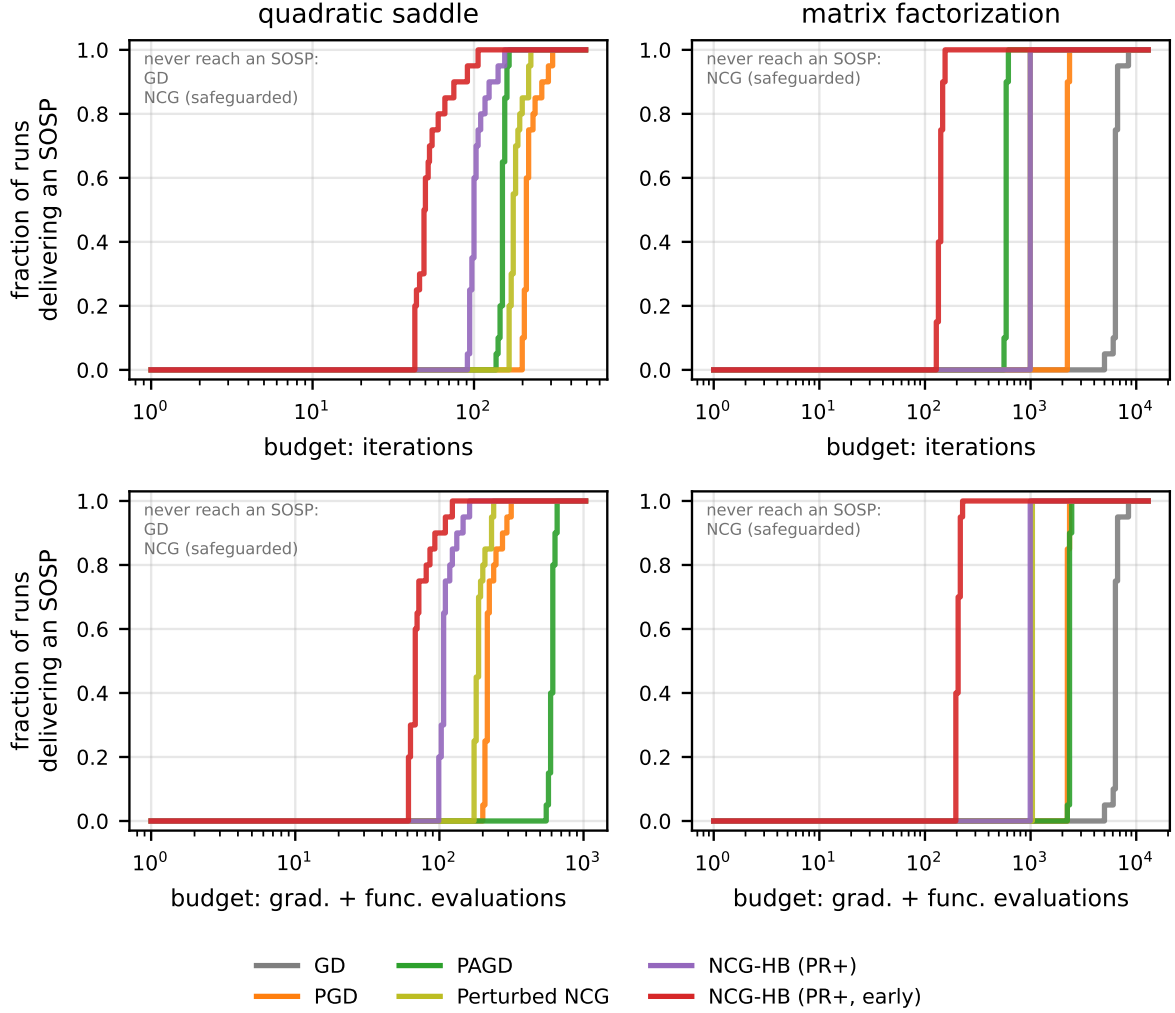


Figure 4.3: Fraction of the 20 repetitions that have reached an $(\varepsilon_g, \varepsilon_H)$ -SOSP within a given budget of iterations (top) and of oracle evaluations (bottom). Methods absent from a panel never reach one on that instance and are named in the corner.

4.5 Which design choices matter

Window length The window length T given by the ledger is much longer than the escape needs, so it is natural to ask how far it can be cut before the method starts returning points that are not SOSPs. Figure 4.4 replaces T by τT for $\tau = 1/16, \dots, 1$ and counts, over 20 repetitions, how often the returned point is a genuine SOSP.

At $\tau = 1/16$ every repetition returns a strict saddle, on both instances. As τ grows the method becomes correct again, but not at the same point for the two problems: matrix factorization is correct 16 times out of 20 at $\tau = 1/8$ and on all 20 repetitions at $\tau = 1/4$, while the quadratic saddle is correct only 4 times out of 20 at $\tau = 1/8$ and 17 out of 20 at $\tau = 1/4$, and

needs $\tau = 1/2$ to be reliable. Hence, the window can safely be shortened by a factor of two to four, so most of the certified budget is unused; but the safe factor is not the same on the two instances, and below it the method fails far more often than the 10% we allow.

Shortening the window to the smallest safe value is faster: the run then certifies and stops after 214 iterations instead of 422 on the quadratic saddle ($\tau = 1/2$), and 504 instead of 1909 on matrix factorization ($\tau = 1/4$). The early exit variant is improved as well, with 152 iterations for the quadratic saddle, and 372 for matrix factorization. We report this to show how much slack the ledger carries, not as something a user could do: finding the smallest safe τ requires knowing when a returned point is a strict saddle, which requires the Hessian, the same objection that applies to ε_H in (4.5). Shortening T while leaving F_{cert} at its original value also breaks the identity (3.44) that relates the two.

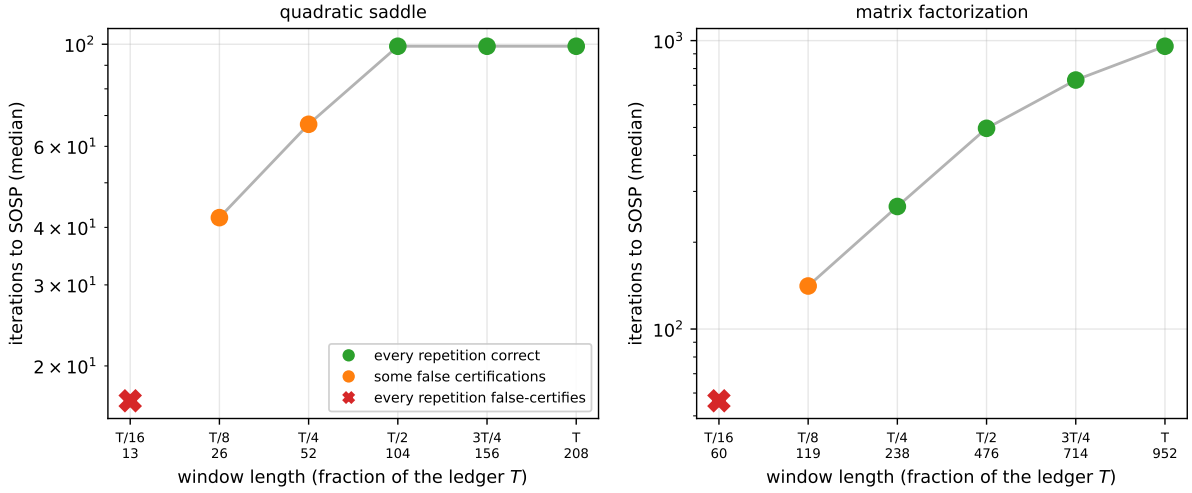


Figure 4.4: Iterations to the SOSP as the window budget is cut to a fraction τ of each instance’s T (208 and 952; the absolute window length is given under each tick), over 20 repetitions. Green: every repetition certifies a true SOSP; orange: some false certifications; red cross: every repetition false-certifies.

Window subroutine The escape times of Figure 4.1 satisfy $\text{HB} \approx \text{AGD} < \text{GD}$ on both instances, with no failures. At the constant momentum $\gamma < \frac{1}{2}$, the accelerated window is not faster than HB and both are faster than GD. This is the empirical content of Remark 6, and the reason the window is left as HB everywhere else.

Against a negative-curvature oracle Replacing the window by an explicit query of the smallest Hessian eigenpair, that is building a NCG/NCD hybrid similar to the algorithm of section 9.3.1 of [62], escapes each saddle in a single step: 11 outer iterations against 99 on the quadratic saddle and 23 against 956 on matrix factorization, using only two and three eigendecompositions per run.

This is the exchange Algorithm 2 makes. As Section 2.2 records, the oracle is invoked at every candidate stationary point, and even in its cheapest form each invocation costs several Hessian-vector products or gradient differences. The HB window attains the same $(\varepsilon_g, \varepsilon_H)$ guarantee with no second-order access of any kind, and pays for it in outer iterations.

4.6 Limitations

The numerical experiments still suffer limitations. The runs use the constants of (4.6) with (3.45) dropped, so they are not certified by Theorem 1. The curvature tolerance (4.5) is derived from the known saddle curvature of each instance, which a practitioner does not have, and neither is the smallest reliable window fraction of Section 4.5. The escape experiments start at a saddle, so they measure the mechanism once a saddle is met and say nothing about how often that happens. On those starts the runs are almost entirely window steps. Table 4.3 therefore compares mostly escape behaviour, and the evidence on the descent phase rests on Section 4.3 alone. Finally, two instances, both with planted structure, are a narrow base from which to generalize.

Chapter 5

Local Convergence under the Strict Saddle Property

So far the target has been an $(\varepsilon_g, \varepsilon_H)$ -SOSP, the standard objective in nonconvex optimization. Without further structure nothing forces it to be a local minimizer. In this chapter we add one landscape assumption, the strict saddle property of [16], under which a certified SOSP is guaranteed to lie near a local minimizer, in a region where f is strongly convex. A safeguarded NCG phase run from that SOSP then converges to the minimizer, with an explicit rate.

Section 5.1 shows that under the strict saddle property the point returned by Algorithm 2 lies in a strongly convex basin. Section 5.2 defines the local phase (Algorithm 3), which continues from the point returned by Algorithm 2 with safeguarded NCG. Section 5.3 counts the iterations of each phases and Section 5.4 sums it all in the final complexity.

5.1 Setting

Definition 1 (Strict Saddle Property, [16]). f is $(\zeta_{ss}, \beta_{ss}, \gamma_{ss}, \delta_{ss})$ -strict saddle if for every x , at least one of the following holds:

1. **Large gradient** $\|\nabla f(x)\| \geq \zeta_{ss}$,
2. **Negative curvature** $\lambda_{\min}(\nabla^2 f(x)) \leq -\beta_{ss}$,
3. **Near a local minimizer** there is a local minimizer x^* with $\|x - x^*\| \leq \delta_{ss}$ and f is γ_{ss} -strongly convex on $B_{x^*}(2\delta_{ss})$.

The point of running Algorithm 2 with tolerances below the strict-saddle thresholds is that it rules out the first two regims described above, leaving only the third.

Lemma 18. *Let f be $(\zeta_{ss}, \beta_{ss}, \gamma_{ss}, \delta_{ss})$ -strict saddle and run Algorithm 2 with $\varepsilon_g < \zeta_{ss}$ and $\varepsilon_H < \beta_{ss}$. On the success event of Theorem 1, which has probability at least $1 - \delta_{\text{tot}}$, the returned point $\tilde{x}$ lies within δ_{ss} of a local minimizer x^* on whose neighbourhood $B_{x^*}(2\delta_{ss})$ the function f is γ_{ss} -strongly convex.*

Proof. The point returned by Theorem 1 has $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \geq -\varepsilon_H$. Since $\varepsilon_g < \zeta_{ss}$, case (1) of Definition 1 fails at $\tilde{x}$, and since $\varepsilon_H < \beta_{ss}$, so does case (2). The three cases being exhaustive, case (3) holds at $\tilde{x}$. $\square$

We write k_0 for the index of the returned point, so that $x_{k_0} = \tilde{x}$. Let

$$\Delta_k := f(x_k) - f(x^*). \quad (5.1)$$

For $x_k \in B_{x^*}(2\delta_{\text{ss}})$, smoothness and γ_{ss} -strong convexity give the standard inequalities (using $\nabla f(x^*) = 0$):

$$\begin{aligned} \frac{\gamma_{\text{ss}}}{2} \|x_k - x^*\|^2 &\leq \Delta_k \leq \frac{L_1}{2} \|x_k - x^*\|^2, \\ 2\gamma_{\text{ss}} \Delta_k &\leq \|g_k\|^2 \leq 2L_1 \Delta_k, \end{aligned} \quad (5.2)$$

where the left half of the second line is the Polyak–Łojasiewicz inequality.

At entry in the strongly convex basin, $\|x_{k_0} - x^*\| \leq \delta_{\text{ss}}$ and (5.2) give

$$\Delta_{k_0} \leq \frac{L_1 \delta_{\text{ss}}^2}{2}. \quad (5.3)$$

Since Δ_k is nonincreasing (Lemmas 1 and 2), the right half of the second line of (5.2) together with (5.3) gives $\|g_k\|^2 \leq 2L_1 \Delta_k \leq 2L_1 \Delta_{k_0} \leq L_1^2 \delta_{\text{ss}}^2$, that is

$$\|g_k\| \leq L_1 \delta_{\text{ss}} \quad \text{for all } k \geq k_0 \text{ with } x_k \in B_{x^*}(2\delta_{\text{ss}}). \quad (5.4)$$

5.2 The Local Phase

Let us call the local phase the safeguarded NCG run from the result of Algorithm 2. Let $\varepsilon_{\text{loc}} \in (0, \varepsilon_g)$ be the tolerance at which the local phase stops, that is when $\|g_k\| \leq \varepsilon_{\text{loc}}$.

Algorithm 3 PRNCG-HB-L: Algorithm 2 followed by local refinement

```

1: Input: the inputs of Algorithm 2, with  $\varepsilon_g < \zeta_{\text{ss}}$  and  $\varepsilon_H < \beta_{\text{ss}}$ ; local tolerance  $\varepsilon_{\text{loc}} \in (0, \varepsilon_g)$ .
2: Run Algorithm 2; let  $\tilde{x}$  be the point it returns.
3:  $x_{k_0} \leftarrow \tilde{x}$ ;  $g_{k_0} \leftarrow \nabla f(x_{k_0})$ ;  $d_{k_0} \leftarrow -g_{k_0}$  ▷ fresh direction
4: for  $k = k_0, k_0 + 1, \dots$  do
5:   if  $\|g_k\| \leq \varepsilon_{\text{loc}}$  then
6:     return  $x_k$  ▷ Theorem 2
7:   end if
8:    $\alpha_k \leftarrow \theta^{j_k}$ ,  $j_k$  minimal with  $f(x_k + \theta^{j_k} d_k) \leq f(x_k) + \eta \theta^{j_k} g_k^\top d_k$ 
9:    $x_{k+1} \leftarrow x_k + \alpha_k d_k$ ;  $g_{k+1} \leftarrow \nabla f(x_{k+1})$ 
10:   $d_{k+1}^{\text{trial}} \leftarrow -g_{k+1} + \beta_{k+1} d_k$ 
11:  if (2.6) holds for  $d_{k+1}^{\text{trial}}$  then
12:     $d_{k+1} \leftarrow d_{k+1}^{\text{trial}}$ 
13:  else
14:     $d_{k+1} \leftarrow -g_{k+1}$ 
15:  end if
16: end for

```

The analysis below targets $\Delta_k \leq \varepsilon_{\text{loc}}^2 / (2L_1)$, which by the right half of (5.2) implies the algorithmic stopping test $\|g_k\| \leq \varepsilon_{\text{loc}}$; bounding the number of iterations to reach the former therefore bounds the running time of Algorithm 3.

For the rest of the chapter we abbreviate

$$a_p := 1 + p, \quad b_p := 2(1 + p - q), \quad r := \frac{\max\{a_p, b_p\}}{2}, \quad (5.5)$$

where p and q are the restart parameters of (2.6).

For $k \in \mathcal{R}$, combining (3.4) with the Polyak–Łojasiewicz inequality of (5.2) turns the per-step decrease into a contraction,

$$\Delta_{k+1} \leq (1 - 2\gamma_{\text{ss}} c_{\mathcal{R}}) \Delta_k, \quad (5.6)$$

so restart steps converge linearly on their own. For $k \in \mathcal{N}$, (3.3) and (5.5) give only

$$\Delta_k - \Delta_{k+1} \geq c_{\mathcal{N}} \min \{ \|g_k\|^{a_p}, \|g_k\|^{b_p} \},$$

whose behaviour depends on the exponents and is analysed in Section 5.3.

5.3 Iteration counts

Restart steps are counted directly from (5.6). Non-restart steps need one preliminary observation: the decrease guarantee (3.3) involves $\min\{\|g_k\|^{a_p}, \|g_k\|^{b_p}\}$, and which of the two exponents is active depends on whether the gradient exceeds one. Since $t \mapsto t^s$ is increasing in s for $t \geq 1$ and decreasing in s for $0 < t \leq 1$,

$$\min(\|g_k\|^{a_p}, \|g_k\|^{b_p}) = \begin{cases} \|g_k\|^{\min(a_p, b_p)} & \text{if } \|g_k\| \geq 1, \\ \|g_k\|^{\max(a_p, b_p)} = \|g_k\|^{2r} & \text{if } \|g_k\| \leq 1. \end{cases} \quad (5.7)$$

We therefore count non-restart steps in two groups, according to whether $\|g_k\| \geq 1$ or $\|g_k\| \leq 1$, writing K_{loc}^H and K_{loc}^L for the two counts. The first group is normally empty, as the following remark shows.

Remark 7. The local phase is entered at the certified point $\tilde{x}$, where $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$. The Polyak–Łojasiewicz inequality (5.2) gives the entry bound $\Delta_{k_0} \leq \varepsilon_g^2/(2\gamma_{\text{ss}})$, sharper than the positional bound (5.3) whenever $\varepsilon_g \leq \sqrt{\gamma_{\text{ss}} L_1} \delta_{\text{ss}}$. Since Δ_k is nonincreasing, smoothness then gives $\|g_k\| \leq \sqrt{2L_1 \Delta_{k_0}} \leq \varepsilon_g \sqrt{L_1/\gamma_{\text{ss}}}$ at every local-phase iterate. Consequently, if $\varepsilon_g \leq \sqrt{\gamma_{\text{ss}}/L_1}$, no local iterate has $\|g_k\| \geq 1$ and $K_{\text{loc}}^H = 0$. We state the bound in full nonetheless, since ε_g is a free tolerance and the condition need not hold.

5.3.1 Restart steps

Lemma 19. *Suppose $x_k \in B_{x^*}(2\delta_{\text{ss}})$ for every local-phase iterate and $2\gamma_{\text{ss}}c_{\mathcal{R}} < 1$. Then the number of restart steps performed before $\Delta_k \leq \varepsilon_{\text{loc}}^2/(2L_1)$ is reached satisfies*

$$K_{\text{loc}}^{\mathcal{R}} \leq \left\lceil \frac{1}{2\gamma_{\text{ss}}c_{\mathcal{R}}} \log \left(\frac{L_1^2 \delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right) \right\rceil. \quad (5.8)$$

Proof. Write $\theta_{\mathcal{R}} := 1 - 2\gamma_{\text{ss}}c_{\mathcal{R}} \in (0, 1)$ and let $k_0 \leq \tau_1 < \tau_2 < \dots$ be the restart steps of the local phase. By (5.6), $\Delta_{\tau_i+1} \leq \theta_{\mathcal{R}} \Delta_{\tau_i}$, and every other local-phase step decreases Δ_k as well (Lemmas 1 and 2), so

$$\Delta_{\tau_K+1} \leq \theta_{\mathcal{R}}^K \Delta_{k_0}.$$

Suppose the stopping condition has not been met at step $\tau_K + 1$, that is $\Delta_{\tau_K+1} > \varepsilon_{\text{loc}}^2/(2L_1)$. Then $\theta_{\mathcal{R}}^K \Delta_{k_0} > \varepsilon_{\text{loc}}^2/(2L_1)$, and taking logarithms,

$$K \log \theta_{\mathcal{R}} > \log \left(\frac{\varepsilon_{\text{loc}}^2}{2L_1 \Delta_{k_0}} \right).$$

Dividing by $\log \theta_{\mathcal{R}} < 0$ reverses the inequality,

$$K < \frac{1}{-\log \theta_{\mathcal{R}}} \log \left(\frac{2L_1 \Delta_{k_0}}{\varepsilon_{\text{loc}}^2} \right),$$

and since $-\log(1-x) \geq x$ on $[0, 1)$ gives $1/(-\log \theta_{\mathcal{R}}) \leq 1/(2\gamma_{\text{ss}}c_{\mathcal{R}})$, the entry bound (5.3) yields

$$K < \frac{1}{2\gamma_{\text{ss}}c_{\mathcal{R}}} \log \left(\frac{2L_1 \Delta_{k_0}}{\varepsilon_{\text{loc}}^2} \right) \leq \frac{1}{2\gamma_{\text{ss}}c_{\mathcal{R}}} \log \left(\frac{L_1^2 \delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right).$$

Hence the number of restart steps before the stopping condition is met is at most the ceiling of the right-hand side, which is (5.8). $\square$

5.3.2 Large-gradient steps

Lemma 20. *Suppose $x_k \in B_{x^*}(2\delta_{\text{ss}})$ for every local-phase iterate and $q \leq 1 + p$. Then*

$$K_{\text{loc}}^H \leq \left\lceil \frac{L_1 \delta_{\text{ss}}^2}{2c_{\mathcal{N}}} \right\rceil. \quad (5.9)$$

Proof. By (3.3) and (5.7), a step with $\|g_k\| \geq 1$ satisfies $\Delta_k - \Delta_{k+1} \geq c_{\mathcal{N}} \|g_k\|^{\min(a_p, b_p)}$. Since $q \leq 1 + p$ gives $b_p \geq 0$, we have $\min(a_p, b_p) \geq 0$ and hence $\|g_k\|^{\min(a_p, b_p)} \geq 1$: each such step decreases Δ by at least $c_{\mathcal{N}}$. Summing over these steps, and using that every other local-phase step also decreases Δ together with the entry bound (5.3),

$$\frac{L_1 \delta_{\text{ss}}^2}{2} \geq \Delta_{k_0} \geq \sum_{k: \|g_k\| \geq 1} (\Delta_k - \Delta_{k+1}) \geq K_{\text{loc}}^H c_{\mathcal{N}}.$$

□

The restriction $q \leq 1 + p$ is the standing assumption of Theorem 1, so it costs nothing here. Outside it the same argument applies with $\|g_k\|^{b_p} \geq (L_1 \delta_{\text{ss}})^{b_p}$ from (5.4) in place of $\|g_k\|^{\min(a_p, b_p)} \geq 1$, at the price of a factor $(L_1 \delta_{\text{ss}})^{2(q-1-p)}$.

5.3.3 Small-gradient steps

For a step with $\|g_k\| \leq 1$, (3.3), (5.7) and the Polyak–Łojasiewicz inequality of (5.2) give

$$\Delta_k - \Delta_{k+1} \geq c_{\mathcal{N}} \|g_k\|^{2r} \geq c_{\mathcal{N}} (2\gamma_{\text{ss}} \Delta_k)^r,$$

that is,

$$\Delta_{k+1} \leq (1 - c_{\mathcal{N}} (2\gamma_{\text{ss}})^r \Delta_k^{r-1}) \Delta_k \quad \text{when } \|g_k\| \leq 1. \quad (5.10)$$

The count K_{loc}^L splits into three subcases according to r . Throughout, k_1 denotes the first iteration with $\|g_k\| \leq 1$; by monotonicity, $\Delta_{k_1} \leq \Delta_{k_0} \leq L_1 \delta_{\text{ss}}^2/2$.

Subcase L1: $r = 1$ (linear convergence). This corresponds to $p = 1$ and $q \geq 1$, or $p < 1$ and $q = p$.

Lemma 21. *Suppose $x_k \in B_{x^*}(2\delta_{\text{ss}})$ for every local-phase iterate, $r = 1$, and $2c_{\mathcal{N}}\gamma_{\text{ss}} < 1$. Then*

$$K_{\text{loc}}^L \leq \left\lceil \frac{1}{2c_{\mathcal{N}}\gamma_{\text{ss}}} \log \left(\frac{L_1^2 \delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right) \right\rceil. \quad (5.11)$$

Proof. Write $\theta_L := 1 - 2c_{\mathcal{N}}\gamma_{\text{ss}} \in (0, 1)$ and let $k_1 \leq \tau_1 < \tau_2 < \dots$ be the steps with $\|g_k\| \leq 1$. With $r = 1$, (5.10) gives $\Delta_{\tau_i+1} \leq \theta_L \Delta_{\tau_i}$, and every other local-phase step decreases Δ_k , so

$$\Delta_{\tau_K+1} \leq \theta_L^K \Delta_{k_1}.$$

Suppose the stopping condition has not been met at step $\tau_K + 1$, that is $\Delta_{\tau_K+1} > \varepsilon_{\text{loc}}^2/(2L_1)$. Taking logarithms and dividing by $\log \theta_L < 0$,

$$K < \frac{1}{-\log \theta_L} \log \left(\frac{2L_1 \Delta_{k_1}}{\varepsilon_{\text{loc}}^2} \right) \leq \frac{1}{2c_{\mathcal{N}}\gamma_{\text{ss}}} \log \left(\frac{L_1^2 \delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right),$$

using $-\log(1-x) \geq x$ on $[0, 1)$ for the second inequality and $\Delta_{k_1} \leq L_1 \delta_{\text{ss}}^2/2$ for the third. Hence the number of such steps is at most the ceiling of the right-hand side. □

Subcase L2: $r < 1$ (finitely many non-restart steps). This corresponds to $p < 1$ and $q > p$.

Lemma 22. *Suppose $x_k \in B_{x^*}(2\delta_{ss})$ for every local-phase iterate and $r < 1$. Then*

$$K_{\text{loc}}^L \leq \left\lceil \frac{(L_1 \delta_{ss}^2/2)^{1-r}}{(1-r) c_{\mathcal{N}} (2\gamma_{ss})^r} \right\rceil. \quad (5.12)$$

Proof. Write $c_k := c_{\mathcal{N}}(2\gamma_{ss})^r \Delta_k^{r-1}$, so that (5.10) reads $\Delta_{k+1} \leq (1 - c_k) \Delta_k$.

If $c_k \geq 1$ at some such step, then $\Delta_{k+1} \leq 0$, and since $\Delta_{k+1} \geq 0$ we get $\Delta_{k+1} = 0$: the target is met and no further such step occurs. We may therefore assume $c_k \in (0, 1)$ at every step counted below.

Define $\varphi_k := \Delta_k^{1-r}$. Since $1 - r \in (0, 1)$, the map $t \mapsto t^{1-r}$ is increasing in t on $(0, \infty)$, so raising $\Delta_{k+1} \leq (1 - c_k) \Delta_k$ to the power $1 - r$ preserves the inequality, and the concavity bound $(1 - c)^{1-r} \leq 1 - (1 - r)c$ for $c \in (0, 1)$ gives

$$\varphi_{k+1} \leq (1 - (1 - r)c_k) \varphi_k = \varphi_k - (1 - r) c_{\mathcal{N}} (2\gamma_{ss})^r,$$

where the equality uses $c_k \varphi_k = c_{\mathcal{N}}(2\gamma_{ss})^r \Delta_k^{r-1} \Delta_k^{1-r} = c_{\mathcal{N}}(2\gamma_{ss})^r$.

Let $k_1 \leq \tau_1 < \tau_2 < \dots$ denote the steps with $\|g_k\| \leq 1$. On every other local-phase step Δ_k decreases, and $1 - r > 0$, so $\varphi_k = \Delta_k^{1-r}$ does not increase there either. Telescoping over this subsequence therefore gives

$$\varphi_{\tau_K+1} \leq \varphi_{k_1} - (1 - r) c_{\mathcal{N}} (2\gamma_{ss})^r K,$$

and since $\varphi_{\tau_K+1} \geq 0$,

$$K_{\text{loc}}^L \leq \frac{\varphi_{k_1}}{(1 - r) c_{\mathcal{N}} (2\gamma_{ss})^r} \leq \frac{(L_1 \delta_{ss}^2/2)^{1-r}}{(1 - r) c_{\mathcal{N}} (2\gamma_{ss})^r},$$

using $\varphi_{k_1} = \Delta_{k_1}^{1-r} \leq (L_1 \delta_{ss}^2/2)^{1-r}$. □

Remark 8. For $r < 1$ the number of non-restart steps with $\|g_k\| \leq 1$ is a constant independent of ε_{loc} , but this does not make the local phase $\mathcal{O}(1)$: by Lemma 3, for $q > p$ the safeguard is unsatisfiable below the gradient threshold $(\sigma/\kappa)^{1/(q-p)}$, so near x^* every step is a restart, and the local complexity is governed by the linear rate of Lemma 19: $\mathcal{O}(\log(1/\varepsilon_{\text{loc}}))$.

Subcase L3: $r > 1$ (sublinear convergence). This corresponds to $p > 1$ or $q < p$.

Lemma 23. *Suppose $x_k \in B_{x^*}(2\delta_{ss})$ for every local-phase iterate, $r > 1$, and $c_{\mathcal{N}}(2\gamma_{ss})^r \Delta_{k_1}^{r-1} < 1$. Then*

$$K_{\text{loc}}^L \leq \left\lceil \frac{(2L_1)^{r-1}}{(r-1) c_{\mathcal{N}} (2\gamma_{ss})^r \varepsilon_{\text{loc}}^{2(r-1)}} \right\rceil. \quad (5.13)$$

Proof. Write $c_k := c_{\mathcal{N}}(2\gamma_{ss})^r \Delta_k^{r-1}$. Since $r - 1 > 0$ and Δ_k is nonincreasing, $c_k \leq c_{k_1} < 1$ by hypothesis at every such step.

Define $\varphi_k := \Delta_k^{1-r}$; since $1 - r < 0$, the map $t \mapsto t^{1-r}$ is decreasing in t on $(0, \infty)$, so raising $\Delta_{k+1} \leq (1 - c_k) \Delta_k$ to the power $1 - r$ reverses the inequality, and the convexity bound $(1 - c)^{1-r} \geq 1 - (1 - r)c$ for $c \in (0, 1)$ gives

$$\varphi_{k+1} \geq (1 - (1 - r)c_k) \varphi_k = \varphi_k + (r - 1) c_{\mathcal{N}} (2\gamma_{ss})^r,$$

using $c_k \varphi_k = c_{\mathcal{N}}(2\gamma_{ss})^r$ as before.

Let $k_1 \leq \tau_1 < \tau_2 < \dots$ denote the steps with $\|g_k\| \leq 1$. On every other local-phase step Δ_k decreases, and $1 - r < 0$, so $\varphi_k = \Delta_k^{1-r}$ does not decrease there either. Telescoping over this subsequence gives

$$\varphi_{\tau_K+1} \geq \varphi_{k_1} + (r-1) c_{\mathcal{N}} (2\gamma_{\text{ss}})^r K.$$

Since $1 - r < 0$, the stopping condition $\Delta_k \leq \varepsilon_{\text{loc}}^2 / (2L_1)$ is equivalent to $\varphi_k \geq (2L_1 / \varepsilon_{\text{loc}}^2)^{r-1}$, so it is met at step $\tau_K + 1$ as soon as

$$\varphi_{k_1} + (r-1) c_{\mathcal{N}} (2\gamma_{\text{ss}})^r K \geq \left(\frac{2L_1}{\varepsilon_{\text{loc}}^2} \right)^{r-1}.$$

Dropping $\varphi_{k_1} \geq 0$ yields (5.13). $\square$

5.4 Total complexity

The local-phase iteration count splits into restart and non-restart contributions, the latter split according to whether $\|g_k\| \geq 1$ or $\|g_k\| \leq 1$:

$$K_{\text{loc}} = K_{\text{loc}}^{\mathcal{R}} + K_{\text{loc}}^H + K_{\text{loc}}^L.$$

Theorem 2. Let f be $(\zeta_{\text{ss}}, \beta_{\text{ss}}, \gamma_{\text{ss}}, \delta_{\text{ss}})$ -strict saddle and let $\varepsilon_{\text{loc}} \in (0, \varepsilon_g)$. Run Algorithm 3 with $\varepsilon_g < \zeta_{\text{ss}}$ and $\varepsilon_H < \beta_{\text{ss}}$, and suppose that

- (i) $q \leq 1 + p$;
- (ii) the local-phase iterates remain in $B_{x^*}(2\delta_{\text{ss}})$;
- (iii) $2\gamma_{\text{ss}}c_{\mathcal{R}} < 1$; and, where applicable, $2c_{\mathcal{N}}\gamma_{\text{ss}} < 1$ (case $r = 1$) and $c_{\mathcal{N}}(2\gamma_{\text{ss}})^r (L_1\delta_{\text{ss}}^2/2)^{r-1} < 1$ (case $r > 1$), the latter implying the condition of Lemma 23 at Δ_{k_1} by (5.3).

Then, on the success event of Theorem 1, the local phase reaches $\Delta_k \leq \varepsilon_{\text{loc}}^2 / (2L_1)$, and hence $\|\nabla f(x_k)\| \leq \varepsilon_{\text{loc}}$, within

$$K_{\text{loc}} \leq \underbrace{\left\lceil \frac{1}{2\gamma_{\text{ss}}c_{\mathcal{R}}} \log \left(\frac{L_1^2\delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right) \right\rceil}_{K_{\text{loc}}^{\mathcal{R}}} + \underbrace{\left\lceil \frac{L_1\delta_{\text{ss}}^2}{2c_{\mathcal{N}}} \right\rceil}_{K_{\text{loc}}^H} + \underbrace{\begin{cases} \left\lceil \frac{1}{2c_{\mathcal{N}}\gamma_{\text{ss}}} \log \left(\frac{L_1^2\delta_{\text{ss}}^2}{\varepsilon_{\text{loc}}^2} \right) \right\rceil & \text{if } r = 1, \\ \left\lceil \frac{(L_1\delta_{\text{ss}}^2/2)^{1-r}}{(1-r)c_{\mathcal{N}}(2\gamma_{\text{ss}})^r} \right\rceil & \text{if } r < 1, \\ \left\lceil \frac{(2L_1)^{r-1}}{(r-1)c_{\mathcal{N}}(2\gamma_{\text{ss}})^r \varepsilon_{\text{loc}}^{2(r-1)}} \right\rceil & \text{if } r > 1, \end{cases}}_{K_{\text{loc}}^L} \quad (5.14)$$

where r is as in (5.5).

Proof. Every local-phase iteration is a restart step or a non-restart step, and every non-restart step has $\|g_k\| \geq 1$ or $\|g_k\| \leq 1$ by (5.7). The three counts are bounded by Lemma 19, by Lemma 20, valid under hypothesis (i), and by whichever of Lemmas 21 to 23 applies according to r . Summing gives (5.14). $\square$

Remark 9. Hypothesis (ii) assumes that the local-phase iterates stay in $B_{x^*}(2\delta_{\text{ss}})$, where f is strongly convex. We do not know how to prove it. Lemma 18 puts x_{k_0} within δ_{ss} of x^* and every step decreases f , but neither bounds the length of a single step, so an iterate could leave the basin. A proof would need control of f along the search segments and not only at the iterates, which the Armijo rule started from $\alpha = 1$ does not provide.

Corollary 1. *Under the hypotheses of Theorem 2, as $\varepsilon_{\text{loc}} \rightarrow 0$ with the strict-saddle parameters $(\zeta_{\text{ss}}, \beta_{\text{ss}}, \gamma_{\text{ss}}, \delta_{\text{ss}})$ held fixed,*

$$K_{\text{loc}} = \begin{cases} O(\log(1/\varepsilon_{\text{loc}})) & \text{if } r \leq 1, \\ O(\varepsilon_{\text{loc}}^{-2(r-1)}) & \text{if } r > 1, \end{cases} \quad (5.15)$$

with constants depending on $L_1, \gamma_{\text{ss}}, c_{\mathcal{R}}, c_{\mathcal{N}}, \delta_{\text{ss}}$ and r but not on ε_{loc} .

Remark 10. Remark 8 settles the case $q > p$. For $q = p$ the safeguard imposes no threshold on the gradient scale, and how often restarts occur near x^* depends on the algorithm and its parameters. This does not affect (5.15): in that regime restart and non-restart steps contract at comparable linear rates (Lemma 19 and Subcase L1), so the order of the bound is the same whichever type dominates.

Corollary 2. *Let Assumptions 1 to 3 hold, let f be $(\zeta_{\text{ss}}, \beta_{\text{ss}}, \gamma_{\text{ss}}, \delta_{\text{ss}})$ -strict saddle, and let the hypotheses of Theorems 1 and 2 hold with $\varepsilon_g < \zeta_{\text{ss}}$, $\varepsilon_H < \beta_{\text{ss}}$ and $\varepsilon_{\text{loc}} \in (0, \varepsilon_g)$. Then, with probability at least $1 - \delta_{\text{tot}}$, Algorithm 3 outputs a point x_k satisfying*

$$\|x_k - x^*\| \leq \frac{\varepsilon_{\text{loc}}}{\sqrt{\gamma_{\text{ss}} L_1}}$$

for some local minimizer x^ of f , after at most $N + K_{\text{loc}}$ iterations, with N as in Theorem 1 and K_{loc} as in Theorem 2.*

Proof. On the success event of Theorem 1, which has probability at least $1 - \delta_{\text{tot}}$, Lemma 18 places the returned point inside the γ_{ss} -strongly convex basin, so the local phase starts there and Theorem 2 applies on the same event. It gives $\Delta_k \leq \varepsilon_{\text{loc}}^2 / (2L_1)$ within K_{loc} iterations, and the first line of (5.2) then yields

$$\|x_k - x^*\| \leq \sqrt{\frac{2\Delta_k}{\gamma_{\text{ss}}}} \leq \sqrt{\frac{\varepsilon_{\text{loc}}^2}{\gamma_{\text{ss}} L_1}} = \frac{\varepsilon_{\text{loc}}}{\sqrt{\gamma_{\text{ss}} L_1}}.$$

The iteration count is the sum of the N iterations of Algorithm 2 and the K_{loc} iterations of the local phase. $\square$

Chapter 6

Conclusion

In this thesis, we derived a second-order complexity guarantee for an NCG-based method, a class for which no such guarantee was previously available. The method alternates a safeguarded restarted NCG phase with a perturbed HB escape phase, and returns an $(\varepsilon_g, \varepsilon_H)$ -SOSP with high probability under Lipschitz gradient and Hessian. The escape phase uses gradient evaluations only, and replaces the negative-curvature oracle that hybrid methods of Section 2.2 invoke at every candidate stationary point. The worst-case bound matches that of PGD, though it does not reach the best known complexity result achieved by PAGD. This gap appears due to the proof technique that we use (see Remark 6). Still, our numerical experiments show that our method can outperform other perturbed first-order techniques on strict saddle instances. Choosing the various constants appearing in the algorithm (and the proof) is key to the method's performance.

6.1 Short-term perspectives

Does NCG escape saddle points? This work does not answer that question, since NCG is used only in the descent phase. Doing so would mean running it inside the escape window, and the current proof technique needs both runs to take the same steps: a line search and a nonlinearly computed momentum coefficient break that, and neither can currently be controlled in terms of the gap between the two trajectories. Letting the momentum coefficient vary, while keeping it below $\frac{1}{2}$, is a smaller relaxation and a natural place to start.

Which pairs of methods work best? Since the two phases are independent, any method with a per-step decrease can occupy the descent phase, and any perturbed method that escapes saddle points can occupy the escape phase. This work analyses one pair, but many others could be analyzed using the same techniques. Which descent methods pair well with which escape mechanisms, and whether the choice should depend on the problem class, remains open.

6.2 Long-term perspective

The work conducted during this master thesis is intended to be submitted for publication in a scientific journal. It is meant to represent the first project of a PhD thesis, that I will start following the internship under supervision of Clément Royer. The PhD thesis will broadly focus on exploiting landscape properties in nonconvex optimization algorithms, with applications in scientific imaging and inverse problems.

Appendix A

Technical lemmas: powers of 2×2 matrices

Lemma 24. *Let $B \in \mathbb{C}^{2 \times 2}$ with eigenvalues μ_1, μ_2 and spectral radius $\rho = \max\{|\mu_1|, |\mu_2|\}$. If $\mu_1 \neq \mu_2$, then for every $m \geq 1$,*

$$\|B^m\| \leq C(B) \rho^m, \quad C(B) := \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|}. \quad (\text{A.1})$$

The bound also holds at $m = 0$ whenever $C(B) \geq 1$.

Proof. Since B is 2×2 , the Cayley–Hamilton theorem lets us write every power as an affine combination of B and I ,

$$B^m = a_m B + b_m I,$$

and evaluating this identity on each eigenvalue gives the scalar system $\mu_i^m = a_m \mu_i + b_m$, $i = 1, 2$. As $\mu_1 \neq \mu_2$ it solves to

$$a_m = \frac{\mu_1^m - \mu_2^m}{\mu_1 - \mu_2}, \quad b_m = \frac{\mu_1 \mu_2^m - \mu_2 \mu_1^m}{\mu_1 - \mu_2}.$$

Bounding numerators termwise, $|\mu_1^m - \mu_2^m| \leq 2\rho^m$ and $|\mu_1 \mu_2^m - \mu_2 \mu_1^m| \leq 2\rho^{m+1}$, so

$$\|B^m\| \leq |a_m| \|B\| + |b_m| \leq \frac{2\rho^m}{|\mu_1 - \mu_2|} \|B\| + \frac{2\rho^{m+1}}{|\mu_1 - \mu_2|} = \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|} \rho^m.$$

At $m = 0$ the left-hand side is $\|I\| = 1$ and the right-hand side is $C(B)$, so the bound holds as soon as $C(B) \geq 1$. $\square$

Lemma 25. *Let $B \in \mathbb{C}^{2 \times 2}$ with eigenvalues μ_1, μ_2 and spectral radius $\rho = \max\{|\mu_1|, |\mu_2|\}$. Let $T = \begin{pmatrix} \mu_1 & s \\ 0 & \mu_2 \end{pmatrix}$. Then for every $m \geq 1$,*

$$T^m = \begin{pmatrix} \mu_1^m & c_m \\ 0 & \mu_2^m \end{pmatrix}, \quad c_m = s \sum_{j=0}^{m-1} \mu_1^j \mu_2^{m-1-j}, \quad |c_m| \leq |s| m \rho^{m-1}. \quad (\text{A.2})$$

If moreover $B = QTQ^*$ is a Schur form of B with Q unitary, then $|s| \leq \|B\|_F$ and, for every $m \geq 1$,

$$\|B^m\| \leq \sqrt{2} \rho^m + \|B\|_F m \rho^{m-1}. \quad (\text{A.3})$$

Proof. Power formula. As a product of upper-triangular matrices T^m is upper triangular, and its diagonal is (μ_1^m, μ_2^m) ; only the $(1, 2)$ entry c_m requires computation. Reading the $(1, 2)$ entry of $T^m = T T^{m-1}$ gives the recurrence

$$c_m = \mu_1 c_{m-1} + s \mu_2^{m-1}, \quad c_0 = 0.$$

We check by induction that $c_m = s \sum_{j=0}^{m-1} \mu_1^j \mu_2^{m-1-j}$. It holds at $m = 1$, where $c_1 = s$. If it holds at $m - 1$, then reindexing gives

$$\mu_1 c_{m-1} = s \sum_{j=1}^{m-1} \mu_1^j \mu_2^{m-1-j},$$

and adding $s\mu_2^{m-1}$, the $j = 0$ term, restores the full sum. Each of the m summands has modulus at most ρ^{m-1} , hence $|c_m| \leq |s| m \rho^{m-1}$. When $\mu_1 = \mu_2 = \mu$ the sum collapses to $m\mu^{m-1}$, recovering the Jordan superdiagonal.

Norm bound. The Frobenius norm is unitarily invariant: $\|QAQ^*\|_F^2 = \text{tr}(QA^*Q^*QAQ^*) = \text{tr}(A^*A) = \|A\|_F^2$. Hence $\|T\|_F = \|B\|_F$, and since

$$\|T\|_F^2 = |\mu_1|^2 + |s|^2 + |\mu_2|^2 \geq |s|^2,$$

we get $|s| \leq \|B\|_F$. The spectral norm is likewise unitarily invariant, so $\|B^m\| = \|T^m\|$. Bounding it by the Frobenius norm and using (A.2),

$$\|B^m\| \leq \|T^m\|_F = \sqrt{|\mu_1|^{2m} + |\mu_2|^{2m} + |c_m|^2},$$

and by $|\mu_1|^{2m} + |\mu_2|^{2m} \leq 2\rho^{2m}$, $|c_m| \leq |s| m \rho^{m-1}$ and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$,

$$\|B^m\| \leq \sqrt{2}\rho^m + |s| m \rho^{m-1} \leq \sqrt{2}\rho^m + \|B\|_F m \rho^{m-1}.$$

□

Bibliography

- [1] Zeyuan Allen-Zhu and Yuanzhi Li. Neon2: Finding local minima via first-order oracles. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.
- [2] Animashree Anandkumar and Rong Ge. Efficient approaches for escaping higher order saddle points in non-convex optimization. In Vitaly Feldman, Alexander Rakhlin, and Ohad Shamir, editors, *29th Annual Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 81–102, Columbia University, New York, New York, USA, 23–26 Jun 2016. PMLR.
- [3] Srinadh Bhojanapalli, Behnam Neyshabur, and Nati Srebro. Global optimality of local search for low rank matrix recovery. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016.
- [4] Emmanuel J. Candès, Xiaodong Li, and Mahdi Soltanolkotabi. Phase retrieval via wirtinger flow: Theory and algorithms. *IEEE Transactions on Information Theory*, 61(4):1985–2007, 2015.
- [5] Coralie Cartis, Nicholas I. M. Gould, and Philippe L. Toint. Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results. *Mathematical Programming*, 127(2):245–295, April 2011.
- [6] Coralie Cartis, Nicholas I. M. Gould, and Philippe L. Toint. Optimal Newton-type methods for nonconvex optimization. Technical Report naXys-17-2011, Dept of Mathematics, FUNDP, Namur (B), 2011.
- [7] Rémi Chan-Renous-Legoubin and Clément W. Royer. A nonlinear conjugate gradient method with complexity guarantees and its application to nonconvex regression. *EURO Journal on Computational Optimization*, 10:100044, 2022.
- [8] Andrew R. Conn, Nicholas I.M. Gould, and Philippe L. Toint. *Trust Region Methods*. MOS-SIAM Series on Optimization. Society for Industrial and Applied Mathematics, 2000.
- [9] Christopher Criscitiello and Nicolas Boumal. Efficiently escaping saddle points on manifolds. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems 32*, pages 5985–5995. Curran Associates, Inc., 2019.
- [10] Frank E. Curtis, Daniel P. Robinson, Clément W. Royer, and Stephen J. Wright. Trust-region newton-cg with strong second-order complexity guarantees for nonconvex optimization. *SIAM Journal on Optimization*, 31(1):518–544, 2021.
- [11] Y. H. Dai. Conjugate gradient methods with Armijo-type line searches. *Acta Mathematicae Applicatae Sinica, English Series*, 18:123–130, 2002.

- [12] Shuvomoy Das Gupta, Robert M. Freund, Xu Andy Sun, and Adrien Taylor. Nonlinear conjugate gradient methods: worst-case convergence rates via computer-assisted analyses. *Mathematical Programming*, 213(1):1–49, September 2025.
- [13] Yann N. Dauphin, Razvan Pascanu, Caglar Gulcehre, Kyunghyun Cho, Surya Ganguli, and Yoshua Bengio. Identifying and attacking the saddle point problem in high-dimensional non-convex optimization. In *Proceedings of the 28th International Conference on Neural Information Processing Systems - Volume 2*, NIPS’14, page 2933–2941, Cambridge, MA, USA, 2014. MIT Press.
- [14] Simon Du, Chi Jin, Jason D Lee, Michael Jordan, Aarti Singh, and Barnabas Póczos. Gradient descent can take exponential time to escape saddle points. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
- [15] R. Fletcher and C. M. Reeves. Function minimization by conjugate gradients. *The Computer Journal*, 7(2):149–154, January 1964.
- [16] Rong Ge, Furong Huang, Chi Jin, and Yang Yuan. Escaping from saddle points - online stochastic gradient for tensor decomposition. In Peter Grünwald, Elad Hazan, and Satyen Kale, editors, *Proceedings of The 28th Conference on Learning Theory, COLT 2015, Paris, France, July 3-6, 2015*, volume 40 of *JMLR Workshop and Conference Proceedings*, pages 797–842. JMLR.org, 2015.
- [17] Rong Ge, Jason D Lee, and Tengyu Ma. Matrix completion has no spurious local minimum. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016.
- [18] Jean Charles Gilbert and Jorge Nocedal. Global convergence properties of conjugate gradient methods for optimization. *SIAM Journal on Optimization*, 2(1):21–42, 1992.
- [19] Baptiste Goujaud, Adrien Taylor, and Aymeric Dieuleveut. Provable non-accelerations of the heavy-ball method. *Mathematical Programming*, October 2025.
- [20] Florentin Goyens and Clément W. Royer. Riemannian trust-region methods for strict saddle functions with complexity guarantees. *Mathematical Programming*, 213(1):863–905, September 2025.
- [21] L. Grippo and S. Lucidi. A globally convergent version of the Polak-Ribière conjugate gradient method. *Mathematical Programming*, 78(3):375–391, September 1997.
- [22] L. Grippo and S. Lucidi. Convergence conditions, line search algorithms and trust region implementations for the polak-ribière conjugate gradient method. *Optimization Methods and Software*, 20(1):71–98, 2005.
- [23] William Hager. A survey of nonlinear conjugate gradient method. *Pacific Journal of Optimization*, 2, 01 2006.
- [24] William W. Hager and Hongchao Zhang. A new conjugate gradient method with guaranteed descent and an efficient line search. *SIAM Journal on Optimization*, 16(1):170–192, 2005.
- [25] William W. Hager and Hongchao Zhang. Algorithm 851: CG_DESCENT, a conjugate gradient method with guaranteed descent. *ACM Trans. Math. Softw.*, 32(1):113–137, March 2006.

- [26] Moritz Hardt. Understanding alternating minimization for matrix completion. In *Proceedings of the 2014 IEEE 55th Annual Symposium on Foundations of Computer Science, FOCS '14*, page 651–660, USA, 2014. IEEE Computer Society.
- [27] Thomas Y. Hou, Zhenzhen Li, and Ziyun Zhang. Analysis of asymptotic escape of strict saddle sets in manifold optimization. *SIAM Journal on Mathematics of Data Science*, 2(3):840–871, 2020.
- [28] Ya-Ping Hsieh, Mohammad Reza Karimi Jaghargh, Andreas Krause, and Panayotis Mertikopoulos. Riemannian stochastic optimization methods avoid strict saddle points. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 29580–29601. Curran Associates, Inc., 2023.
- [29] Chi Jin, Rong Ge, Praneeth Netrapalli, Sham M. Kakade, and Michael I. Jordan. How to escape saddle points efficiently. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1724–1732. PMLR, 06–11 Aug 2017.
- [30] Chi Jin, Praneeth Netrapalli, and Michael I. Jordan. Accelerated gradient descent escapes saddle points faster than gradient descent. In Sébastien Bubeck, Vianney Perchet, and Philippe Rigollet, editors, *Proceedings of the 31st Conference On Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 1042–1085. PMLR, 06–09 Jul 2018.
- [31] Sahar Karimi and Stephen Vavasis. A single potential governing convergence of conjugate gradient, accelerated gradient and geometric descent, 2019.
- [32] Sahar Karimi and Stephen Vavasis. A single potential governing convergence of conjugate gradient, accelerated gradient and geometric descent, 2019.
- [33] Sahar Karimi and Stephen A. Vavasis. Nonlinear conjugate gradient for smooth convex functions. *Mathematical Programming Computation*, 16(2):229–254, June 2024.
- [34] Kenji Kawaguchi. Deep learning without poor local minima. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016.
- [35] J. Kuczyński and H. Woźniakowski. Estimating the largest eigenvalue by the power and lanczos algorithms with a random start. *SIAM Journal on Matrix Analysis and Applications*, 13(4):1094–1122, 1992.
- [36] Jason D. Lee, Ioannis Panageas, Georgios Piliouras, Max Simchowitz, Michael I. Jordan, and Benjamin Recht. First-order methods almost always avoid strict saddle points. *Mathematical Programming*, 176(1–2):311–337, 2019.
- [37] Jason D. Lee, Max Simchowitz, Michael I. Jordan, and Benjamin Recht. Gradient descent only converges to minimizers. In Vitaly Feldman, Alexander Rakhlin, and Ohad Shamir, editors, *29th Annual Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 1246–1257, Columbia University, New York, New York, USA, 23–26 Jun 2016. PMLR.
- [38] Huan Li and Zhouchen Lin. Restarted nonconvex accelerated gradient descent: No more polylogarithmic factor in the $o(\epsilon^{-7/4})$ complexity. *Journal of Machine Learning Research*, 24(146):1–49, 2023.

- [39] Jianhao Ma, Geyu Liang, and Salar Fattahi. Understanding the implicit regularization of gradient descent in over-parameterized models, 2025.
- [40] Andreea-Alexandra Muşat and Nicolas Boumal. Gradient descent avoids strict saddles with a simple line-search method too, 2025.
- [41] Andreea-Alexandra Muşat and Nicolas Boumal. A non-autonomous center-stable set theorem for saddle avoidance in optimization, 2026.
- [42] Yurii Nesterov. A method for solving the convex programming problem with convergence rate $o(1/k^2)$. *Proceedings of the USSR Academy of Sciences*, 269:543–547, 1983.
- [43] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*, volume 87 of *Applied Optimization*. Springer, Boston, MA, 2004.
- [44] Yurii Nesterov and Boris Polyak. Cubic regularization of newton method and its global performance. *Math. Program.*, 108:177–205, 08 2006.
- [45] Praneeth Netrapalli, U N Niranjan, Sujay Sanghavi, Animashree Anandkumar, and Prateek Jain. Non-convex robust pca. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014.
- [46] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, New York, NY, 2nd edition, 2006.
- [47] Peter Ochs, Yunjin Chen, Thomas Brox, and Thomas Pock. ipiano: Inertial proximal algorithm for nonconvex optimization. *SIAM Journal on Imaging Sciences*, 7(2):1388–1419, 2014.
- [48] Michael O’Neill and Stephen J. Wright. Behavior of accelerated gradient methods near critical points of nonconvex functions. *Mathematical Programming*, 176(1):403–427, July 2019.
- [49] Ioannis Panageas and Georgios Piliouras. Gradient descent only converges to minimizers: Non-isolated critical points and invariant regions. In *8th Innovations in Theoretical Computer Science Conference (ITCS 2017)*, volume 67 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 2:1–2:12. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2017.
- [50] Ioannis Panageas, Georgios Piliouras, and Xiao Wang. First-order methods almost always avoid saddle points: The case of vanishing step-sizes. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.
- [51] Barak A. Pearlmutter. Fast exact multiplication by the hessian. *Neural Computation*, 6(1):147–160, 1994.
- [52] Polak, E. and Ribiere, G. Note sur la convergence de méthodes de directions conjuguées. *R.I.R.O.*, 3(16):35–43, 1969.
- [53] B.T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964.
- [54] Radoslaw Pytlak. *Conjugate Gradient Algorithms in Nonconvex Optimization*, volume 89 of *Nonconvex Optimization and Its Applications*. Springer-Verlag, Berlin Heidelberg, 2009.

- [55] Clément W. Royer and Stephen J. Wright. Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization. *SIAM Journal on Optimization*, 28(2):1448–1477, 2018.
- [56] Clément W. Royer, Michael O’Neill, and Stephen J. Wright. A Newton-CG algorithm with complexity guarantees for smooth unconstrained optimization. *Mathematical Programming*, 180(1):451–488, March 2020.
- [57] Trond Steihaug. The conjugate gradient method and trust regions in large scale optimization. *SIAM Journal on Numerical Analysis*, 20(3):626–637, 1983.
- [58] Ju Sun, Qing Qu, and John Wright. Complete dictionary recovery over the sphere i: Overview and the geometric picture. *IEEE Transactions on Information Theory*, 63(2):853–884, February 2017.
- [59] Ju Sun, Qing Qu, and John Wright. A geometric analysis of phase retrieval. *Foundations of Computational Mathematics*, 18(5):1131–1198, August 2017.
- [60] Tao Sun, Dongsheng Li, Zhe Quan, Hao Jiang, Shengguo Li, and Yong Dou. Heavy-ball algorithms always escape saddle points. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI-19*, pages 3520–3526. IJCAI Organization, 2019.
- [61] Yue Sun, Nicolas Flammarion, and Maryam Fazel. Escaping from saddle points on riemannian manifolds. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.
- [62] John Wright and Yi Ma. *High-Dimensional Data Analysis with Low-Dimensional Models: Principles, Computation, and Applications*. Cambridge University Press, 2022.
- [63] Stephen J. Wright and Benjamin Recht. *Gradient Methods Using Momentum*, page 55–74. Cambridge University Press, 2022.
- [64] Yi Xu, Rong Jin, and Tianbao Yang. First-order stochastic algorithms for escaping from saddle points in almost linear time. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.
- [65] S. K. Zavriev and F. V. Kostyuk. Heavy-ball method in nonconvex optimization problems. *Computational Mathematics and Modeling*, 4(4):336–341, October 1993.